ARTICLE | Article in Press

# Graph neural networks for antarctic sea ice concentration forecasting

**Tobias Milz · Varvara Vetrova · Marwan Katurji**

Received: 18 May 2026 / Accepted: 7 September 2026
© The Author(s) 2026

**Abstract**

Antarctic sea ice is a crucial component of the climate system, yet in recent years it has undergone abrupt, poorly understood changes that motivate more accurate weeks-ahead forecasts. Existing statistical and dynamical approaches can struggle to represent the strongly nonlinear, heterogeneous, and rapidly evolving nature of Antarctic sea-ice dynamics, while conventional deep-learning models like convolutional neural networks (CNN) typically operate on regular Euclidean grids, making long-range spatial dependencies and irregular regional interactions difficult to represent explicitly. Here we introduce the first graph neural network (GNN) framework for forecasting weekly Antarctic sea ice concentration (SIC) and its anomalies (SICA) on subseasonal time scales of up to eight weeks ahead. Our framework couples an encoder–processor–decoder architecture, trained on satellite observations and ERA5 reanalysis, with an ablation of two contrasting graph-construction schemes: (1) a purely geometric icosahedral mesh, and (2) a data-driven graph built from super-pixel segmentation of the Antarctic domain. In controlled ablations, we found that how the graph is constructed matters more than the choice of message-passing operator (Interaction, GraphSAGE, DeepGAT or DConv). Benchmarked against five re-implemented CNN architectures (IceNet, SICNet-CBAM, SICNet-TSAM, Unicorn, ANTSIC-UNet), two statistical baselines, and the operational ECMWF S2S dynamical forecast, the GNN achieves the lowest error on three out of four metrics, cutting full-grid MAE by roughly 18% relative to the strongest CNN benchmark and by 48% relative to a persistence forecast, while using 4–24 times fewer trainable parameters. Only on root-mean-square error (RMSE) at the ice edge does it fall behind IceNet (by about 4%). To explain our model's prediction, a feature attribution analysis identifies that it relies on sea-surface temperature as the dominant driver beyond the seasonal climatology.

**Keywords** Antarctica · Sea ice concentration · Graph neural networks · ECMWF · Convolutional neural networks · NSIDC

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.





https://doi.org/10.1038/s41598-026-70984-7

Published online: 16 September 2026

scientific reports

## Introduction

Sea ice refers to ice that forms and dissolves in ocean waters. It is a distinct ecosystem that is amongst the largest habitats on the planet, spanning between 3.9 and 4.3% of the Earth's surface[1]. The majority is found in the Southern Hemisphere, where Antarctic sea ice grows from a minimum of 3 million km² in February to about 19 million km² in September — an area larger than the Antarctic continent itself[1].

Its size, seasonal variability, and unique environment make it a crucial component of the climate system. Some of its impacts include the regulation of emission and absorption of $CO_2$[2], the exchange of moisture and energy between the ocean and atmosphere[3], and the salinity composition of the oceans[4]. In a warming climate, changes to sea ice extent and concentration can therefore lead to far-reaching consequences. Among the more immediate and localised effects is the self-reinforcing albedo feedback loop: the loss of reflective sea ice leads to higher sea surface temperatures, which in turn reinforces further sea ice melt[5].

The rapid decline in Arctic sea ice extent has been studied extensively over the last two decades, revealing a clear downward trend since the beginning of satellite records in 1979. While underestimated, climate models did predict this trend due to rising global temperatures[6]. In contrast to climate model projections, however, was the surprising and contradictory increase in Antarctic sea ice extent during the same period[7]. This unexpected trend persisted until 2014, when it abruptly reversed, leading to yet another unforeseen development as a rapid and unprecedented decline unfolded within the next three years. This anomalous decrease in sea ice extent has reached new extremes in recent years (2017, 2022–2025), resulting in the lowest sea ice coverage on record over the last four and a half decades[8,9]. These events highlight that the development of sea ice in the Antarctic is less well understood and exhibits larger seasonal and regional variability[10].

With these rapidly evolving changes comes a growing interest in furthering our understanding of sea ice dynamics and the need for more accurate forecasting of sea ice extent and concentration. Data-driven machine learning (ML) has proven to be a powerful statistical tool, propelling advancements in sea ice-related forecasting over the last decade. While much of the early work was focused on the Arctic, more recent advances have now also been applied to the Antarctic.

Despite the success of these models in terms of competing with dynamical models and statistical baselines, they, for the most part, follow a grid-based convolutional paradigm. Graph neural networks (GNNs) represent a fundamentally different inductive bias from convolutional neural networks (CNNs), because rather than processing data on a regular Euclidean grid, they operate on arbitrary graph topologies, enabling explicit modelling of long-range teleconnections and irregular spatial dependencies. Their efficacy has been demonstrated for weather prediction[11], sea surface temperature forecasting[12], and ENSO modelling[13], yet they have not been tested for the forecasting of sea ice dynamics. This leaves two open questions: First, it remains unknown whether the advantages GNNs have demonstrated in other geoscientific forecasting domains carry over to sea ice prediction, and how such models compare against the established convolutional and dynamical approaches. Second, applying a GNN to gridded geophysical data requires constructing the underlying graph in the first place, which represents an important modelling decision in its own right, whose influence on forecast skill has received little systematic attention.

In this study, we address both gaps. We present the first GNN-based framework for forecasting Antarctic sea ice concentration (SIC) and its anomaly (SICA), together with a comprehensive comparison to recently published deep learning models and an operational dynamical forecast. Trained on satellite-derived sea ice observations from the National Snow and Ice Data Center (NSIDC)[14] and atmospheric and oceanic predictors from the ERA5 reanalysis, our models forecast weekly mean SIC and SICA fields at lead times of up to eight weeks.

Specifically, we: (1) develop a modified GraphCast-style encoder–processor–decoder GNN that operates jointly on the 1° Antarctic pixel grid and a coarse long-range graph, complemented by grid-local message passing, a latent autoregressive rollout over the forecast horizon, and a modular processor supporting interchangeable message-passing operators (Interaction Network[15], GraphSAGE[16], DeepGAT[17], and diffusion convolution

(DConv)[18,19]); (2) create two coarse-graph constructions including a novel data-driven graph derived from climatological clustering of the Antarctic domain via a super-pixel segmentation approach, and the geometry-based icosahedral multi-mesh of GraphCast[11]; (3) conduct controlled ablation studies isolating the effects of graph construction and of the choice of processor operator; (4) benchmark the resulting model against five re-implemented CNN architectures from the recent sea ice prediction literature (IceNet[20], SICNet-CBAM and SICNet-TSAM[21], Unicorn[22], and ANTSIC-UNet[23]) under a unified evaluation protocol; and **(5)** compare all learned models against persistence and climatology baselines and the operational ECMWF S2S dynamical forecast.

The remainder of this paper is organised as follows. Sections "Arctic sea ice prediction" and "Antarctic sea ice prediction" review related work on ML-based sea ice prediction in the Arctic and Antarctic. We then describe the dataset and the two coarse-graph constructions (Sect. "Data") and the GNN architecture (Sect. "Graph neural network architecture"), before presenting the ablation and cross-model evaluation results on both the SIC and SICA prediction tasks, together with a feature-attribution analysis (Sect. "Results"). We close by discussing the implications and directions for future work. Full data, graph, model, and evaluation details required to reproduce the study are provided in the Methods (Sect. "Methods").

## Related work

ML and deep learning models have rapidly advanced the field of sea ice prediction in recent years. However, much of the early work focused on the Arctic. Despite the stark differences between the two regions, lessons learned from research in either are highly relevant to the other. Also transferable are ideas from other environmental data science domains such as weather or sea-surface temperature forecasting. This section is divided into two parts: Section "Arctic sea ice prediction" covers research in the Arctic and Sect. "Antarctic sea ice prediction" covers research in the Antarctic.

### Arctic sea ice prediction

One of the earliest studies into ML-based prediction of SIC was published by[24]. They developed a deep neural network (DNN) model trained on historical SIC data collected from 1978 to 2016, without incorporating any other physical or environmental variables. The performance of the DNN was evaluated against traditional autoregressive (AR) models and a compilation of forecasts from the Sea Ice Prediction Network (SIPN) on a monthly timescale. Their approach outperformed the AR model and achieved comparable accuracy to the SIPN ensemble forecasts. Building on that, Kim et al.[25] developed a CNN model for a similar task of predicting monthly SIC. The performance of their model was evaluated against traditional machine learning approaches, demonstrating superior accuracy and consistency. As predictors, they used SIC, SICA, SST, 2-meter air temperature, forecast albedo and the v-component of wind speed.

Despite the success of these deep learning models, Fritzner et al.[26] also showed in their study that even simple ML approaches can produce significant improvements over costly dynamical models. They evaluated and compared a high-resolution dynamical model with data assimilation and two statistical models, a fully convolutional network and a kNN approach. Using inputs like sea ice concentration, sea surface temperature, and 2-meter air temperature, the models focus on short-term forecasts (1–4 weeks). Both ML methods outperformed persistence forecasts, and notably, even the kNN model outperformed the dynamical model during stable ice conditions.

The next study worth highlighting was presented by Liu et al.[27]. The authors employ a convolutional long short-term memory network (ConvLSTM) trained on weekly data with lead times of up to 6 weeks. As predictors, they used 2-meter temperature, sea level pressure, net surface turbulent and radiation flux, 10-meter zonal and meridional wind and geopotential height at 850 hPa and 500 hPa. A constrained forecast, which assumes perfect knowledge of future predictors, and an operational forecast, which relies solely on historical data, were evaluated. Their models demonstrate superior performance compared to the persistence and climatology predictions.

In the same year, Andersson et al.[20] introduced **IceNet**. Unlike the previous works, the prediction task of this approach is formulated as a classification problem, where SIC values are grouped into three categories: open water (SIC $\leq$ 15%), marginal ice (15% < SIC < 80%), and full ice (SIC $\geq$ 80%). For each grid cell and lead times (6 months), IceNet generates the monthly average probability distribution across these three classes using an ensemble of U-Net models. As input features, the models use a list of variables from 2-meter temperature, SIC, solar radiation and geopotential height. The final output is obtained by averaging the probability distributions from all 25 ensemble members. The results of this approach show an improvement over ECMWF's SEAS5 model. It is the first model that we adopted as one of our CNN benchmarks.

Ren et al.[21] proposed **SICNet**, a U-Net-structured model with a temporal–spatial attention module that predicts daily SIC from SIC alone at lead times of 7–28 days, improving over CNN- and RNN-based models; its two attention variants, **SICNet-CBAM** and **SICNet-TSAM**, form two further benchmarks in our comparison. Building on this work, Ren and Li[28] also introduced $\text{SICNet}_{90}$, extending the horizon to 90 days and investigating how adding oceanic and atmospheric variables such as SST, sea air temperature, and geopotential height affects predictive skill; each configuration showed significant improvement over climatology.

Similar to the previous work by Fritzner et al.[26], Lin et al.[29] introduced Ice-kNN, a computationally efficient model for daily Arctic summer sea ice forecasting with lead times ranging from 1 to 122 days. The enhanced Ice-kNN model outperforms both climatological and anomaly persistence forecasts using predictors from ERA5 data such as 2-meter temperature, sea-level pressure and surface net heat flux.

Most recently, Park et al.[22] introduced **Unicorn**, a U-Net-based model for weekly Arctic SIC prediction with lead times of up to 4 weeks, trained on 12 weeks of input history. Its central architectural contribution is the replacement of the conventional convolutional U-Net bottleneck with a convolutional neural ordinary differential equation (ConvNODE), which evolves the latent state continuously in time and thereby models spatial and temporal dynamics jointly rather than in separate stages. Beyond SIC itself, Unicorn incorporates two ancillary satellite products as inputs: the passive-microwave brightness temperature and sea ice age, the latter serving to distinguish the volatile first-year ice at the margins from the more stable multi-year ice. Across their benchmarks, including the SICNet variants, Unicorn achieved an average MAE improvement of roughly 12% for SIC forecasting and approximately 18% on the integrated ice-edge error for sea ice extent prediction. Unicorn is the fourth CNN-family benchmark re-implemented in this study.

### Antarctic sea ice prediction

One of the first studies was introduced by Wang et al.[30], who developed SIPNet, a spatiotemporal attention-based deep learning model for subseasonal Antarctic SIC prediction. SIPNet utilises a sequence-to-sequence architecture and temporal-spatial attention modules (TSAMs) to capture complex dynamics. As input variables, they rely solely on past SIC values. Their approach was shown to outperform traditional dynamic and statistical models, including ECMWF forecasts and persistence benchmarks for weekly predictions with lead times of 1–8 weeks, the same weekly task formulation we adopt in this study.

Shortly after, Dong et al.[31] employed a ConvLSTM network for daily SIC predictions up to 60 days ahead, again relying only on SIC-derived inputs, and achieved good performance for lead times under 30 days compared to benchmark predictions and a dynamical model from ECMWF.

The next study we want to highlight was conducted by Yang et al.[23], who used a U-Net architecture called **ANTSIC-UNet** to predict monthly SIC at lead times of up to 6 months from 14 input parameters, including 2-metre and upper-air temperature, sea surface temperature, upper-ocean heat content, radiative fluxes, winds, and geopotential height. Their model outperformed a linear-trend model, an anomaly-persistence model, and ECMWF's SEAS5 dynamical forecast system in SIC error and integrated ice-edge error. They additionally quantified predictor importance via feature permutation, finding that for lead times of up to 2 months the model relies mainly on the initial sea ice conditions, surface upward shortwave radiation, sea surface temperature, and the 10-hPa zonal wind. ANTSIC-UNet completes our set of five re-implemented benchmarks.

Similar to their previous model,[32] published (kNN)-South, an adaptation of Ice-kNN. The model is trained to predict daily sea ice concentration anomalies up to 90 days in advance. As input, they again rely solely on information derived from SIC. Their results show that the model outperforms persistence, climatology and ECMWF's dynamic model predictions.

Lastly and most recently, Baek et al.[33] developed a deep learning framework for seasonal prediction of Antarctic sea ice extent anomalies, predicting area-mean values for five Antarctic sectors at a one-season lead rather than gridded concentration fields. Their residual U-Net is pre-trained on the CESM2 large ensemble and then fine-tuned on observations and reanalysis data, using SIC, downward shortwave radiation, near-surface winds, sea level pressure, SST, and subsurface ocean temperature as predictor variables. The model outperforms four operational dynamical forecast systems (ECMWF, DWD, Met Office, and Météo-France).

These pioneering studies above demonstrate the potential of ML in this domain. Yet despite their variety in data preprocessing, spatial and temporal resolutions and models, they share a common design choice: all operate on a regular pixel grid with convolutional (or pixel-wise) architectures. Works from related environmental data science fields such as weather prediction[11,34,35], regional heatwave prediction[36] and sea-surface temperature forecasting[12] have shown that graph-based models offer a promising alternative for modelling complex non-local interactions that might also be relevant to sea-ice evolution. Yet, to the best of our knowledge, no study has applied GNNs and their capacity for explicit long-range, non-Euclidean connectivity to sea ice prediction, nor systematically examined how the underlying graph could be constructed. This represents a notable gap.

## Data

In our experiments, SIC is defined as the fractional coverage of sea ice within a given grid cell, ranging from 0 to 1. This parameter was derived from the NSIDC-0051 dataset, which is produced by passive microwave radiometers (SMMR, SSM/I, and SSMIS) aboard satellite platforms operated by the Defense Meteorological Satellite Program (DMSP)[14]. The spatial coverage ranges from approximately $-39.23°$ to $-90°$ latitude, encompassing the Antarctic region. As the target variables for this study, we used weekly averaged SIC and SICA values. SICA corresponds to the signed deviation of the observed SIC from its climatological expectation for that week as calculated over the training period:

$$\mathrm{SICA}_w = \mathrm{SIC}_w - \overline{\mathrm{SIC}}_w^{\mathrm{clim}} \tag{1}$$

As predictors, we used the ten variables listed in Table 1, the majority taken from ERA5 reanalysis. All data were resampled to a common 1° latitude–longitude grid ($50 \times 360$ cells, 18,000 grid points total) and coarsened from daily to weekly resolution using a non-overlapping 7-day mean. The final dataset comprises 1,777 weekly time steps spanning 1989-01-04 to 2023-01-09. Data were split chronologically into training (70%), validation (15%), and test (15%) sets, with a leakage buffer between partitions (input window plus the forecast horizon) to prevent

**Table 1** List of predictor variables. Input features used for the ML models for the SIC and SICA prediction tasks.

| Predictor Variable | Source Dataset |
|---|---|
| 10 m u-component of wind | ERA5 |
| 10 m v-component of wind | ERA5 |
| 2 m temperature | ERA5 |
| Land-sea mask | ERA5 |
| Mean sea level pressure | ERA5 |
| Sea surface temperature (SST) | ERA5 |
| Total precipitation | ERA5 |
| Sea ice concentration (SIC) | NSIDC |
| SIC anomaly | NSIDC |
| SIC climatology (1989–2012) | NSIDC |

a training setup based on future data. The input context window includes the last four weeks of data for every prediction, resulting in

$$\underbrace{4 \text{ weeks} \times 10 \text{ predictors per week}}_{\text{temporal context}} = \boxed{40 \text{ input variables per grid point}}.$$

Dynamic forcing variables were z-score normalised using training-period statistics while SIC, SICA, and SIC climatology were retained in native physical units. Prediction skill is reported on both the full grid and a **dynamic-ice mask** that retains only those grid cells whose target standard deviation over the training period exceeds 0.01, isolating the marginal ice zone where predictive skill matters most.

## Graphs

Graph neural networks operate on an explicit graph topology rather than a regular pixel lattice, so the choice of graph is itself a modelling decision. We construct and compare two coarse-graph designs as representational forms of the original grid-based data: (1) a purely geometric **icosahedral multi-mesh**, following the GraphCast approach to global weather forecasting[11], and (2) a data-driven **SLIC super-pixel graph** derived from climatological segmentation of the Antarctic circle by means of simple linear iterative clustering (SLIC)[37]. In both cases, the $50 \times 360$ latitude–longitude grid provides the fine-scale **grid nodes** on which inputs and predictions live, while the graph construction supplies a set of **coarse nodes** and three edge sets: (1) encoder edges (grid → graph), (2) node-to-node edges over which the bulk of message passing takes place, and (3) decoder edges (graph → grid).

### Icosahedral multi-mesh

The mesh is initialised as a regular icosahedron, which is the mathematically most uniform placement of 12 points on a sphere. Two nodes are pinned to the geographic poles, and the remaining ten form two rings of five nodes at latitudes of approximately $\pm 26.57°$, with the two rings offset by 36° in longitude so that they interleave. Connecting the nodes along the boundary of their convex hull yields the 20 triangular faces of the initial mesh.

The mesh is then refined $K$ times. Each refinement step splits every triangle into four more refined sub-triangles by inserting the midpoint of each edge and projecting it back onto the unit sphere, giving $V_K = 10 \cdot 4^K + 2$ nodes after $K$ subdivisions. Most importantly, edges of previous levels are **retained** and labelled with the refinement level at which they first appeared. This way, lower-level edges, like the 30 edges of the base icosahedron, still exist even in the refined finer mesh and act as long-range "highways". This multi-scale edge set is the central idea inherited from GraphCast[11], which allows message passing to propagate information locally along fine edges and across the entire domain along coarse edges within a small number of hops.

Figure 1 illustrates this creation process starting from $K = 0$ to $K = 5$ on the globe. Because our domain is the Antarctic, the global mesh is cropped to the spherical cap with latitude $\leq -40°$ (blue dashed line in Fig. 1). Retained nodes are re-indexed, and only multi-mesh edges with both endpoints inside the cap are kept, preserving their level parameter.

Information flow between grid and mesh The grid and mesh are coupled by bipartite edge sets. Each of the 18,000 grid cells is connected to its three nearest mesh nodes (nearest neighbours in three-dimensional Cartesian coordinates), resulting in $18{,}000 \times 3 = 54{,}000$ encoder edges; the decoder edges are these edges reversed. The reasoning behind choosing three encoder edges per grid cell is to add redundancy and ensure better information propagation throughout the graph.

### SLIC super-pixel graph

The SLIC graph is a data-driven alternative in which the coarse nodes are not placed geometrically but derived from the climatological structure of the data itself. SLIC clustering[37], popular in computer vision, identifies

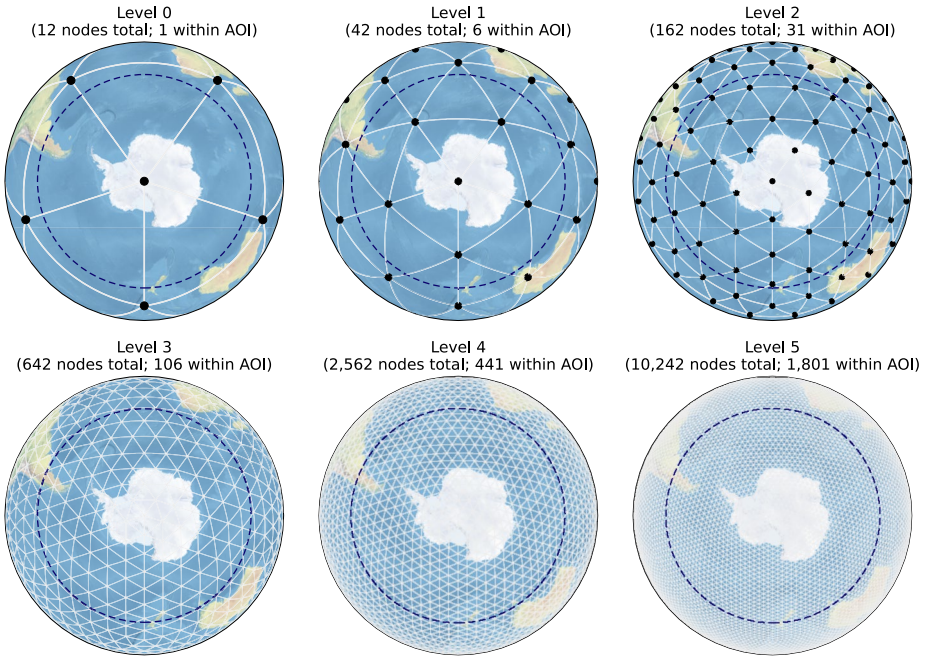



**Fig. 1** The icosahedral multi-mesh at successive refinement levels. Each panel shows one refinement level (0-5). Markers are mesh nodes, arcs are mesh edges; the dashed circle is −40° latitude, the retained Antarctic domain, our area of interest (AOI). Panel titles report total nodes and nodes inside the domain as used in the final graph.

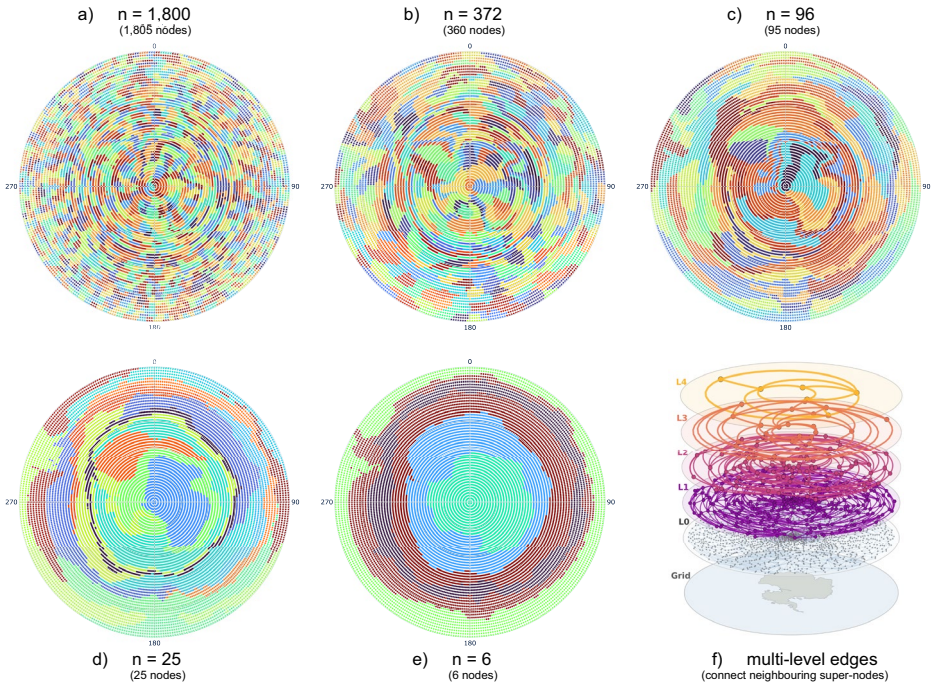



**Fig. 2** SLIC graph creation process in polar stereographic view. Coloured segments in panels (**a**)–(**e**) indicate clusters as identified by the SLIC algorithm given different initialisation conditions (requested number of clusters). Panel (**f**) shows the respective graph connecting the super-nodes of each cluster with their direct neighbours. Higher-level clusters span larger regions, adding longer-ranged connections.

superpixels, segments of coherent or similar patterns. For our data, the segmentation is performed on a single (50, 360, $C$) "image" representing the temporal mean over the training period of the following: sea-ice concentration, 2-meter air temperature, and sea-surface temperature; all common SIC predictors.

Figure 2 illustrates this segmentation at different initialisations. The n-parameter (like the k value in k-Means) gives the algorithm the number of desired regions to segment. The SLIC graph is created by introducing so-called super-nodes, which are nodes at the centre of each of the $\approx n$ segments (or superpixels). They act as aggregative hubs for their representative region and provide information propagation to surrounding areas by connecting to their counterparts of directly neighbouring clusters. To add long-range connections, we run this process

iteratively with consecutive segmentations performed with coarser segment budgets (i.e. smaller $n$ values), thus revealing larger but fewer regions and longer "highway" edges.

This design provides a compact, physically informed hierarchical graph structure that preserves local spatial continuity while facilitating information exchange between distant but dynamically related regions (for further details on the SLIC creation, see Sect. "SLIC super-pixel graph").

Information flow between grid and super-nodes. For this graph, the encoder edges connect every grid cell to the super-node of the region containing it (exactly 18,000 hub edges, one per grid cell independent of the SLIC setting); decoder edges are these reversed.

## Graph neural network architecture

All GNNs in our study follow a **encoder–processor–decoder** architecture operating jointly on the $50 \times 360$ latitude–longitude grid (18,000 grid nodes) and one of the coarse graphs described in the previous section. Each grid node carries the 40 input channels, which are passed to the GNN to produce an 8-week SIC or SICA forecast. Throughout, the models use a hidden width of 128, eight processor layers, SiLU activations, sigmoid (for SIC) and linear (for SICA) output functions, and residual connections with layer normalisation. The architecture of the GNN forecast pipeline is illustrated in Fig. 3. Training and hyperparameter-optimisation details are given in the Methods (Sect. "Training and hyperparameter optimization").

The forward pass proceeds as follows. An input multilayer perceptron (MLP) first embeds the 40 grid channels into a 128-dimensional hidden space. Two **grid-local** message-passing layers then operate among the 18,000 grid nodes themselves, on an 8-connectivity pixel lattice ($\sim$141,544 edges, constructed internally), sharpening local spatial structure at full resolution before any coarsening. The **encoder** pushes the resulting grid states onto the coarse nodes of the selected graph along the encoder edges, after which the **processor** applies eight rounds of message passing, propagating information across the domain. The graph selection as well as the layers of the processor are subject to the ablation studies (see Sect. "Ablation results"); the rest of the architecture is agnostic to these changes. The **decoder** writes the processed coarse-node states back to the grid as a **residual** update using a skip connection from the encoded and refined original input before coarsening; this way, full-resolution detail bypasses the coarse bottleneck akin to a U-Net skip. Two further grid-local layers then refine the decoded state at full resolution. This placement of grid-local layers before the encoder and after the decoder mirrors the local convolutions on the contracting and expanding paths of a U-Net and restores CNN-like local detail at the ice edge, which routing everything through the coarse bottleneck would otherwise smear. Finally, an autoregressive rollout maps each grid node's state to the eight forecast weeks.

However, rather than emitting all eight forecast weeks at once as independent regressions from a single readout MLP, the model evolves each grid node's latent state forward one week at a time with a shared GRU cell[38]. This rollout is autoregressive in **latent** space rather than input space, which means no future forcing variables (e.g. future winds or temperatures) need to be predicted. This latent-evolution strategy parallels Unicorn's ConvNODE backbone[22], which likewise evolves a latent state and decodes one frame per step. The GRU cell takes as input the original input data through a skip connection and concatenates it with the decoded state from the grid-local refinement. Adding this second skip connection allows the model to retain persistence-like local detail.

## Results

In this section, we first present the results of the two ablation studies and evaluate the resulting model against the benchmarks, before repeating the cross-model comparison on the SICA task and analysing parameter and compute efficiency. Throughout, every learned model is trained with five independent random seeds, each evaluation metric is reported as mean ± standard deviation over the seeds and differences are interpreted against the

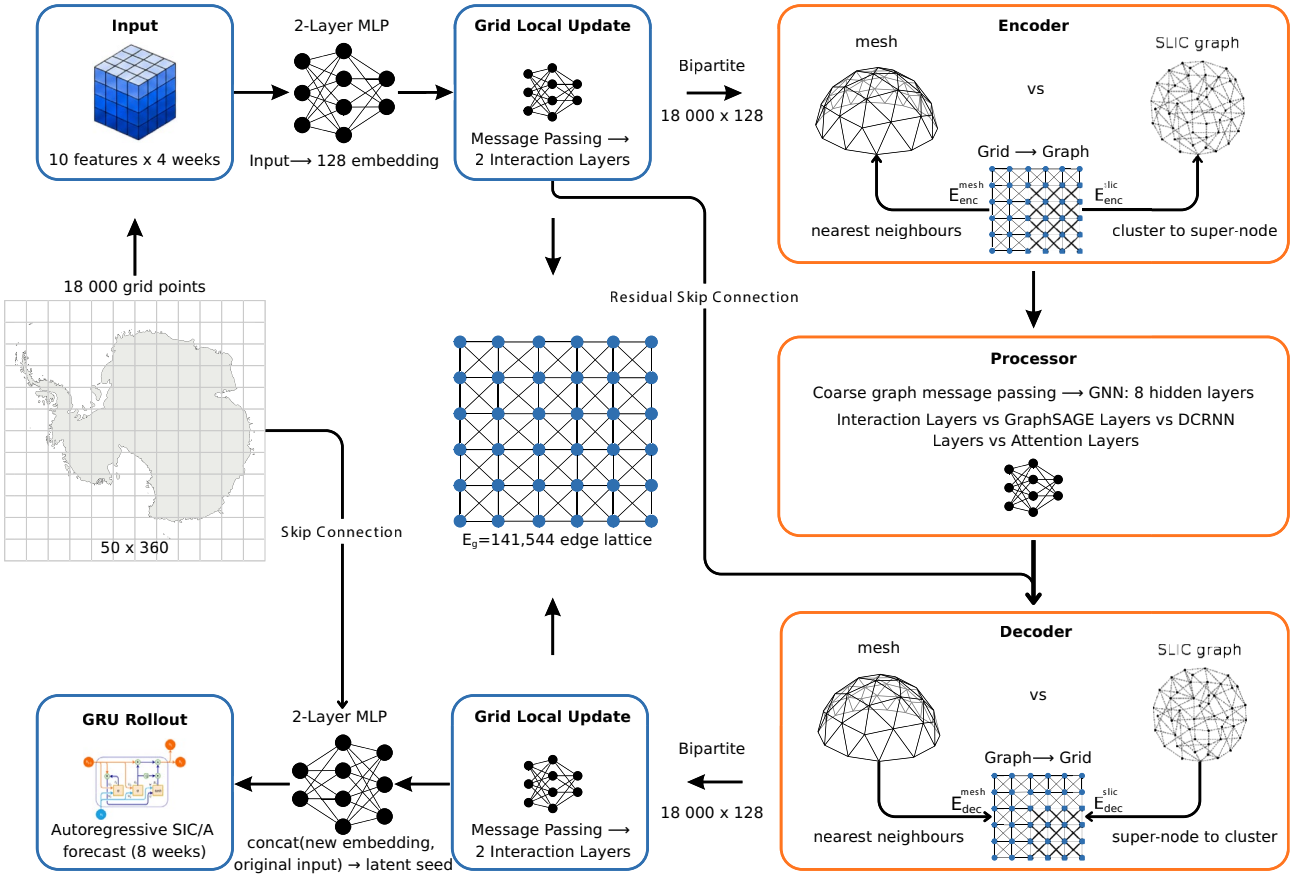



**Fig. 3** GNN SIC/A forecasting pipeline. The figure highlights the pipeline's modularity, which enables ablation studies of the coarse graphs and the different processors. It also illustrates the U-Net-like structure and incorporation of skip connections and autoregression. Blue boxes represent operations performed on the input grid ($G_g$), while the orange elements highlight the encoder–processor–decoder part on the coarse graph ($G_c$).

seed-to-seed spread. All learned models share the same 40 input channels, 8-week output, data split, and are evaluated against MAE and RMSE on the full grid and the dynamic-ice mask over the forecast horizon.

## Ablation results

We isolate two components of the architecture: the coarse graph and the message-passing operator applied within the processor. The operator ablation then fixes the best-performing graph of the graph ablation and swaps only the processor operator.

### Graph construction

The graph ablation contrasts a total of six different configurations that were chosen to balance abstraction against spatial detail relative to the 18,000-cell grid. These six configurations jointly separate two factors: geometric versus data-driven construction, and single-scale versus hierarchical connectivity (see Sect. "Graph ablation" for further details on the selected graph properties). The results of the ablation of these configurations are summarised in Table 2.

The first observation that stands out is that both icosahedral meshes rank above all four SLIC configurations on mean ice-masked RMSE, with the $K$=5 mesh best overall ($0.1121 \pm 0.0010$). Second, increasing the mesh resolution

**Table 2** Graph-construction ablation. The table shows the RMSE results on the dynamic-ice mask, week-averaged over the 8-week horizon, as mean ± standard deviation over five training seeds (lower is better). The last column gives the minimum and maximum seed results. It shows that, while the best-performing seed-run was recorded for the K=6 mesh, on average, it does not improve on the lower K=5 version. Both icosahedral meshes rank above all SLIC configurations, and the hierarchical connectivity consistently improves the SLIC graphs; only the gaps to the flat SLIC graphs exceed the seed-noise band.

| Family | Parameter | Hier. level | Nodes | Edges | En/decoder | RMSE ice (mean ± std) | Seed range |
|---|---|---|---|---|---|---|---|
| Icosahedral mesh | $K = 5$ | 5 | 1, 801 | 6,890 | 54,000 | $\mathbf{0.1121 \pm 0.0010}$ | [0.1108, 0.1129] |
| Icosahedral mesh | $K = 6$ | 6 | 7, 206 | 28,230 | 54,000 | $0.1122 \pm 0.0017$ | [0.1104, 0.1145] |
| SLIC | $n = 7{,}200$ | Hier. (L5) | 4, 486 | 18,095 | 18,000 | $0.1136 \pm 0.0024$ | [0.1113, 0.1168] |
| SLIC | $n = 1{,}800$ | Hier. (L4) | 1, 805 | 6,298 | 18,000 | $0.1142 \pm 0.0021$ | [0.1116, 0.1170] |
| SLIC | $n = 7{,}200$ | Flat (L0) | 4, 486 | 11,863 | 18,000 | $0.1155 \pm 0.0016$ | [0.1142, 0.1181] |
| SLIC | $n = 1{,}800$ | Flat (L0) | 1, 805 | 4,962 | 18,000 | $0.1171 \pm 0.0013$ | [0.1148, 0.1180] |

**Table 3** Processor-operator ablation. RMSE on the dynamic-ice mask, week-averaged over the 8-week horizon, as mean ± standard deviation over five training seeds (lower is better). The coarse graph ($K$=5 mesh) and the surrounding architecture are held fixed; only the mesh-to-mesh operator inside the processor varies. Interaction, GraphSAGE, and DeepGAT are statistically tied; only the diffusion operator underperforms and trains less stably.

| Processor operator | RMSE ice (mean ± std) | Seed range |
|---|---|---|
| DeepGAT | $0.1120 \pm 0.0011$ | [0.1109, 0.1131] |
| Interaction (default) | $0.1121 \pm 0.0010$ | [0.1108, 0.1129] |
| GraphSAGE | $0.1122 \pm 0.0014$ | [0.1106, 0.1144] |
| Diffusion (DConv) | $0.1163 \pm 0.0058$ | [0.1123, 0.1265] |

beyond $K$=5 brings no benefit: the $K$=6 mesh ($0.1122 \pm 0.0017$) differs from $K$=5 by only 0.0001 despite carrying four times as many nodes, so we adopt $K$=5 as the headline graph. Third, within the SLIC family, hierarchy helps consistently as both node budgets of the hierarchical variants outperform their flat, single-scale counterparts by 0.0019 ($n = 7{,}200$) and 0.0029 ($n = 1{,}800$). This indicates that the long-range "highway" edges, rather than the region count, drive most of the difference within the SLIC graphs.

The between-graph spread from best to worst (0.0050) exceeds the mean within-configuration seed standard deviation of 0.0017, so the choice of coarse graph is a real architectural effect rather than training noise. An important qualification, however, is that this effect is predominantly due to the significantly lower performance of the flat SLIC graphs: the margin between the best mesh ($K$=5, 0.1121) and the best hierarchical SLIC graph ($n = 7{,}200$ at L5, 0.1136) is only 0.0015 and lies within the seed-to-seed noise. We thus do not claim a meaningful edge of the mesh over the SLIC approach.

### Message-passing operator

Within the processor we compare four interchangeable message-passing operators: the Interaction Network[15] used by default, in which each message is a learned nonlinear function of the sender state, the receiver state, and the edge's refinement-level embedding; GraphSAGE[16], which aggregates neighbour features by a simple mean; DeepGAT[17], which weights neighbours by learned attention; and diffusion convolution (DConv)[18,19], a bidirectional random-walk diffusion. Interaction and DeepGAT are edge-aware and can read the refinement-level embedding, whereas GraphSAGE and DConv are edge-unaware. Every operator is wrapped in identical residual scaffolding so that only the message and aggregation core differ across the ablation. For the full operator definitions, architectural specifics, and resulting parameter counts, see Sect. "GNN architecture".

Table 3 reports the operator ablation on the best graph from above (the $K$=5 mesh). The results show that the performance of Interaction, GraphSAGE, and DeepGAT are statistically indistinguishable, with seed means of $0.1121 \pm 0.0010$, $0.1122 \pm 0.0014$, and $0.1120 \pm 0.0011$, respectively, which is a span of 0.0002, an order of magnitude below the seed spread. On this graph, the expressiveness of the message function (nonlinear sender–receiver interactions, attention weighting, or plain mean aggregation) is immaterial to forecast skill.

The one exception is the diffusion operator, which is both worse on average ($0.1163 \pm 0.0058$) and markedly less stable across seeds, with one run reaching 0.1265, which is roughly five-fold larger seed standard deviation of any other operator.

Considering these results together with the graph ablation, the graph modelling decision seems to carry significantly more weight than the choice of message-passing operator in the proposed model architecture pipeline.

## SIC forecast results

We evaluate the GNN against three families of reference forecasts: the persistence (tiles the last observed weekly SIC field forward across all eight forecast weeks) and climatology baselines; five convolutional architectures from the recent sea-ice literature (IceNet, SICNet-CBAM, SICNet-TSAM, Unicorn, and ANTSIC-UNet), which we re-implemented and trained from scratch; and the operational ECMWF S2S dynamical forecast. For the latter, the comparison requires separate evaluation concerns as it has to be performed on a smaller test set that matches the available forecast data of the model, which runs only twice a week and has a shorter forecast horizon of 46 days. Furthermore, the outputs had to be interpolated from 1.5 to 1° spatial resolution, adding further caveats to this comparison (for further details on all the benchmark models see Sect. "Benchmarks").

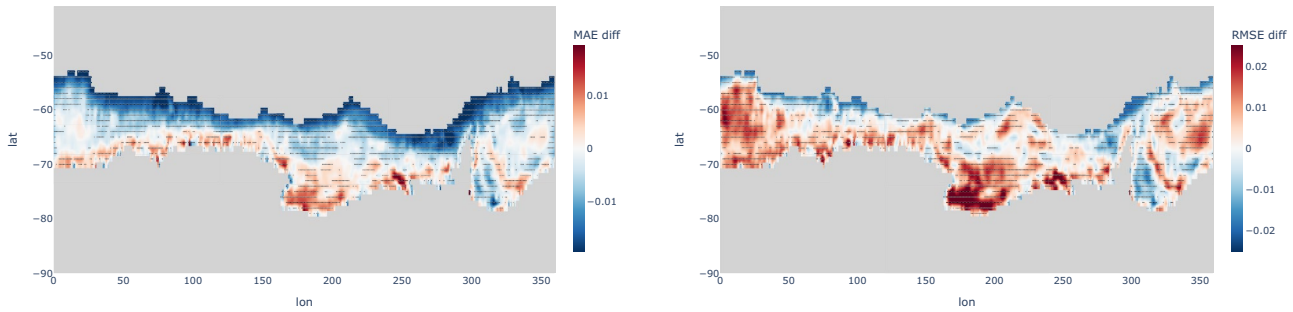
Table 4 lists every model's results separated by model family. The GNNs use the $K$=5 mesh identified as the best-performing graph in the ablation (Sect. "Ablation results"); all scores are week-averaged over the 8-week horizon on the full test set, and we report all four metric–domain combinations plus the full-grid MAE skill relative to both baselines.

The Interaction GNN attains the best score on three of the four metrics. On full-grid MAE it leads by a wide margin ($0.0163 \pm 0.0002$ against 0.0198–0.0315 for the CNNs), corresponding to a 48.0% error reduction relative to persistence and 20.6% relative to climatology. It also leads on ice-masked MAE ($0.0600 \pm 0.0003$) and on full-grid RMSE ($0.0578 \pm 0.0005$), where the DeepGAT variant ties it ($0.0578 \pm 0.0006$). On the fourth metric, RMSE on the dynamic-ice mask, however, IceNet is the best model ($0.1079 \pm 0.0007$ versus the best GNN's 0.1120–0.1121). This gap of roughly 0.004 is well beyond the seed spread of either model and is therefore a genuine difference. The ratio $\frac{RMSE}{MAE}$ is a compact summary of how heavy-tailed a model's error distribution is. For the Interaction GNN, this value is 1.87 against 1.67 for IceNet. Because RMSE penalises large errors disproportionately, the natural reading is that IceNet commits fewer **large** ice-edge misses, even though its **average** error (MAE) is higher on

**Table 4** SIC forecast results. Results are week-averaged over the 8-week forecast horizon (lower is better; best value per column in bold, with the GNN Interaction and DeepGAT variants tied on full-grid RMSE). Neural models are reported as mean ± standard deviation over five training seeds; the statistical baselines are deterministic; "full" denotes the full $50 \times 360$ grid and "ice" the dynamic-ice mask. The skill columns give the reduction in full-grid MAE relative to persistence and climatology, respectively, $100 \times (1 - \mathrm{MAE}_{\mathrm{model}}/\mathrm{MAE}_{\mathrm{baseline}})$. All CNN results are our re-implementations under the common experimental setup, not the originally published numbers.

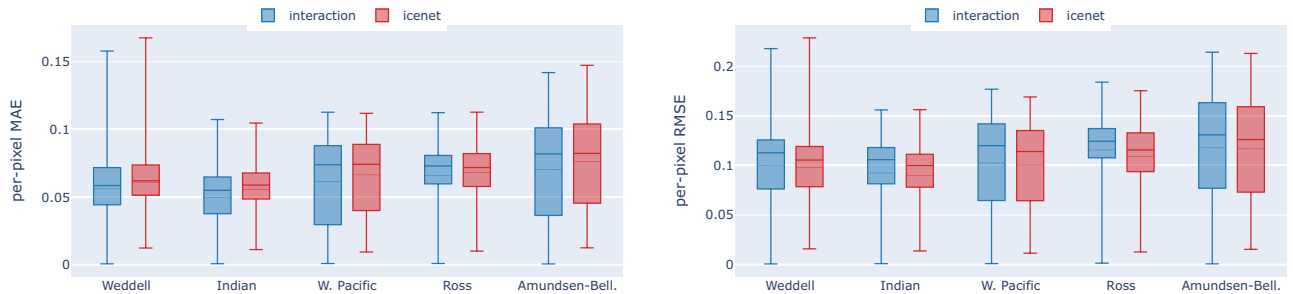
| Model | Family | MAE full | RMSE full | MAE ice | RMSE ice | Skill% (pers.) | Skill% (clim.) |
|---|---|---|---|---|---|---|---|
| Interaction | GNN | **0.0163 ± 0.0002** | **0.0578 ± 0.0005** | **0.0600 ± 0.0003** | 0.1121 ± 0.0010 | **+48.0** | **+20.6** |
| GraphSAGE | GNN | 0.0164 ± 0.0004 | 0.0579 ± 0.0007 | 0.0604 ± 0.0013 | 0.1122 ± 0.0014 | +47.8 | +20.4 |
| DeepGAT | GNN | 0.0165 ± 0.0002 | **0.0578 ± 0.0006** | 0.0604 ± 0.0007 | 0.1120 ± 0.0011 | +47.4 | +19.7 |
| Diffusion (DConv) | GNN | 0.0166 ± 0.0010 | 0.0600 ± 0.0030 | 0.0622 ± 0.0036 | 0.1163 ± 0.0058 | +46.9 | +18.9 |
| IceNet | CNN | 0.0315 ± 0.0008 | 0.0584 ± 0.0004 | 0.0645 ± 0.0013 | **0.1079 ± 0.0007** | −0.4 | −53.3 |
| SICNet-CBAM | CNN | 0.0298 ± 0.0022 | 0.0608 ± 0.0006 | 0.0644 ± 0.0011 | 0.1137 ± 0.0007 | +4.9 | −45.1 |
| SICNet-TSAM | CNN | 0.0314 ± 0.0020 | 0.0611 ± 0.0005 | 0.0648 ± 0.0009 | 0.1133 ± 0.0009 | −0.2 | −53.0 |
| Unicorn | CNN | 0.0198 ± 0.0003 | 0.0592 ± 0.0007 | 0.0618 ± 0.0008 | 0.1143 ± 0.0015 | +36.8 | +3.5 |
| ANTSIC-UNet | CNN | 0.0247 ± 0.0004 | 0.0598 ± 0.0003 | 0.0630 ± 0.0003 | 0.1141 ± 0.0006 | +21.1 | −20.4 |
| Persistence | baseline | 0.0314 | 0.1139 | 0.1178 | 0.2207 | — | — |
| Climatology | baseline | 0.0205 | 0.0715 | 0.0771 | 0.1386 | — | — |

https://doi.org/10.1038/s41598-026-70984-7




(a) Spatial difference map (MAE). (b) Spatial difference map (RMSE).

**Fig. 4** Spatial difference map. Seed-averaged per-pixel error difference at the ice edge (Interaction GNN - IceNet). Blue indicates pixels where Interaction GNN is closer to the observation, red where the CNN is closer. Black stippling marks pixels where the inter-model difference exceeds twice the across-seed standard deviation (robust to initialisation). A pattern emerges showing a non-region-specific outer vs inner ice edge divide.




(a) MAE by region. (b) RMSE by region.

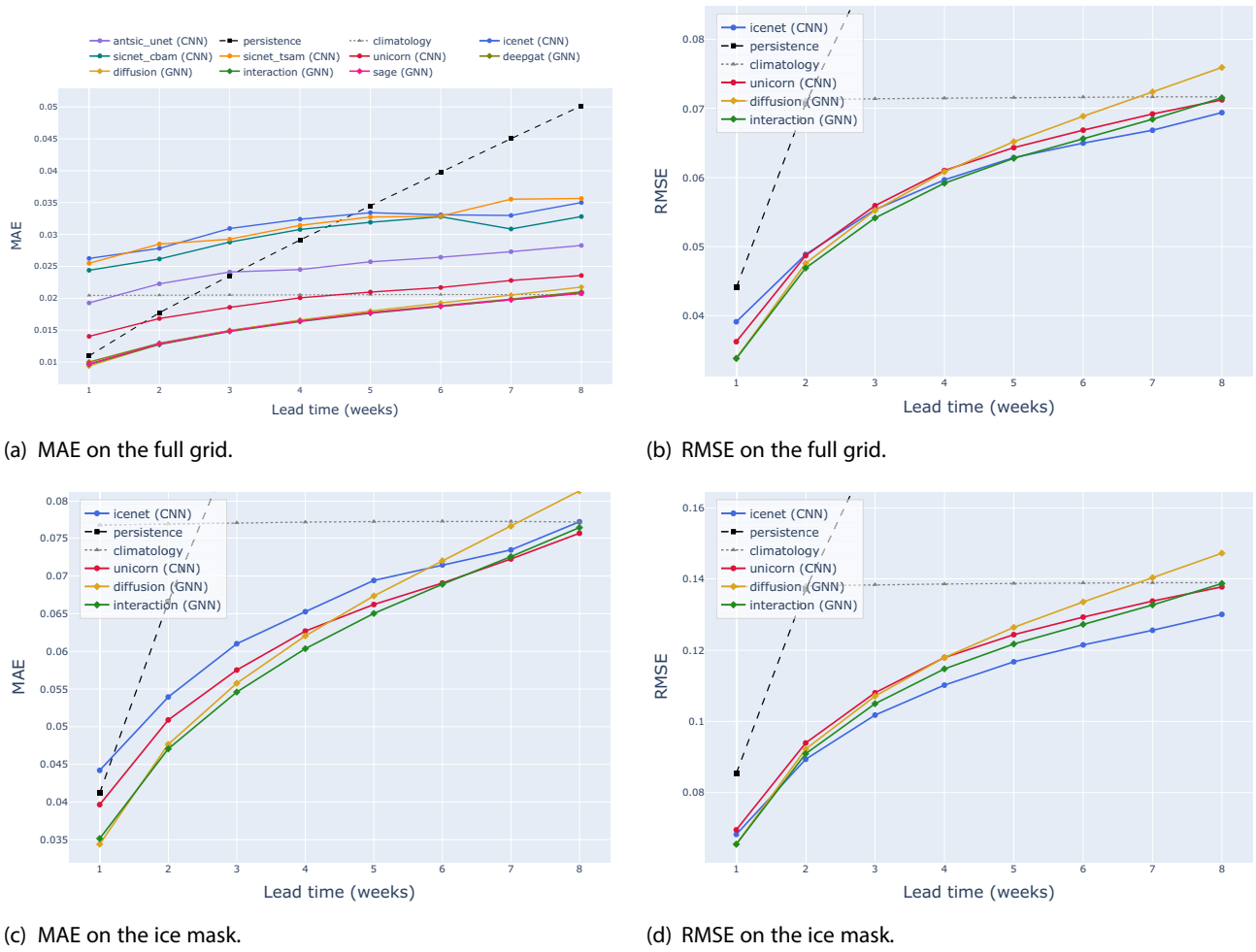
**Fig. 5** Error distribution across Southern Ocean sectors. One box per model per sector (Weddell, Indian, West Pacific, Ross, Amundsen–Bellingshausen). Boxes show the interquartile range (IQR), the central line the median, dotted lines the mean, and whiskers extend to the most extreme values within $1.5 \times$ IQR of the quartiles. The Interaction GNN has the lower MAE and IceNet the lower RMSE in every sector; the overall MAE/RMSE tradeoff is regionally consistent.

both domains. To characterise this tradeoff, we compare the two models directly at the ice edge. Figures 4 and 5 dissect the result spatially:

Figure 4 resolves **where** this difference happens and whether it is robust. Mapping the per-pixel, seed-averaged error difference (Interaction-IceNet) shows the two models trading advantage across the ice edge, with the GNN performing better at the open ice edge and the CNN better at the inner ice edge. Black stippling marks the pixels where the difference exceeds twice the across-seed standard deviation, that is, where the sign of the difference is unlikely to be an artefact of a single lucky or unlucky initialisation. About 3, 100 of the 4, 788 ice-edge pixels are stippled, so most of the inter-model difference is a stable, reproducible effect rather than seed noise.

The box plots in Fig. 5 show that the tradeoff is not a regional artefact either. Broken down by Southern Ocean sector, the Interaction GNN has the lower MAE in every one of the five sectors (see Fig. 5a) and IceNet the lower RMSE in the same (see Fig. 5b). Both models find the Amundsen–Bellingshausen Sea hardest and the Indian Ocean easiest, and the RMSE gap is widest where errors are largest (e.g. Ross: 0.1092 for IceNet versus 0.1156 for the GNN). The MAE-versus-RMSE split is therefore a consistent property of **how** the two architectures differ, holding region and seed-wide.

Figure 6 resolves the horizon-averaged results on the full grid into per-week curves. Figure 6ahighlights the GNN variants' prowess on the MAE full grid metric as they lead over the benchmarks at essentially every lead week, by a large and nearly lead-independent margin: at week 1 the Interaction GNN reaches 0.0096 versus 0.0140 for the best CNN (Unicorn), and only by week 8 does the GNN ($\approx$0.0208) meet the flat climatology level




(a) MAE on the full grid.

(b) RMSE on the full grid.

(c) MAE on the ice mask.

(d) RMSE on the ice mask.

**Fig. 6** SIC forecast results by week. Figure breaks down the results over the forecast horizon. Panels (b)-(d) show the baselines and best-performing ML models only for higher visibility. Panel (a) highlights the GNNs superior performance over the CNNs on the MAE-full grid metric. Furthermore, for every metric, they have the lowest short-range (week 1) error, even for the RMSE on the ice edge where IceNet dominates on average.

(0.0206). Persistence is competitive at week 1 as its full-grid MAE (0.0110) beats every CNN; however, it rapidly collapses with lead. Three CNNs (IceNet, SICNet-TSAM, SICNet-CBAM) also sit above the climatology line at every lead week, explaining their negative averaged skill in Table 4.

Figure 6b–d resolve the other metrics over the forecast horizon, but for visual clarity, they only highlight the baselines and best-performing CNNs and selected GNNs. These more detailed head-to-head comparisons add two nuances that the averaged scores hide. First, the RMSE ranking changes across the horizon: the GNN has the best RMSE at week 1 in both domains, but IceNet overtakes from week 2 at the ice edge and steadily widens its lead through to week 8 (0.1301 versus 0.1387). On the full-grid, IceNet overtakes only after week 6. The GNN is thus the sharpest model at short lead, while IceNet is the most robust at long lead times, most visibly at the ice edge. Second, the diffusion (DConv) variant is the best model of all at week 1 but degrades fastest, ending as the weakest learned model at the ice edge by week 8.

### Comparison with the operational dynamical forecast

As mentioned in Sect. "SIC forecast results", the comparison with the dynamical model had to be conducted on a smaller test set that matches the available forecast data. Table 5 summarises the results of every model on this

**Table 5** SIC forecast results including the operational ECMWF model. This table reads the same as Table 4. However, the models were tested on a subset of the main test set to accommodate the dynamical model's prediction data. The relative performance of the models is consistent with that of the experiment on the full test set. Noteworthy is the superior performance of all ML models compared to the dynamical model.

| Model | Family | MAE full | RMSE full | MAE ice | RMSE ice |
|---|---|---|---|---|---|
| Interaction | GNN | **0.0151 ± 0.0002** | **0.0539 ± 0.0003** | **0.0556 ± 0.0005** | 0.1045 ± 0.0005 |
| GraphSAGE | GNN | 0.0152 ± 0.0003 | 0.0541 ± 0.0006 | 0.0561 ± 0.0011 | 0.1049 ± 0.0011 |
| DeepGAT | GNN | 0.0153 ± 0.0002 | **0.0539 ± 0.0005** | 0.0561 ± 0.0006 | 0.1044 ± 0.0010 |
| Diffusion (DConv) | GNN | 0.0153 ± 0.0006 | 0.0554 ± 0.0019 | 0.0570 ± 0.0022 | 0.1073 ± 0.0037 |
| IceNet | CNN | 0.0307 ± 0.0007 | 0.0554 ± 0.0004 | 0.0612 ± 0.0012 | **0.1021 ± 0.0007** |
| SICNet-CBAM | CNN | 0.0293 ± 0.0023 | 0.0579 ± 0.0006 | 0.0614 ± 0.0011 | 0.1081 ± 0.0008 |
| SICNet-TSAM | CNN | 0.0301 ± 0.0019 | 0.0580 ± 0.0005 | 0.0616 ± 0.0007 | 0.1076 ± 0.0009 |
| Unicorn | CNN | 0.0189 ± 0.0003 | 0.0559 ± 0.0006 | 0.0584 ± 0.0006 | 0.1079 ± 0.0012 |
| ANTSIC-UNet | CNN | 0.0239 ± 0.0003 | 0.0567 ± 0.0003 | 0.0600 ± 0.0004 | 0.1081 ± 0.0007 |
| ECMWF S2S | Dynamical | 0.0372 | 0.1347 | 0.1397 | 0.2612 |
| Persistence | Baseline | 0.0260 | 0.0976 | 0.0975 | 0.1892 |
| Climatology | Baseline | 0.0207 | 0.0717 | 0.0776 | 0.1390 |

**Table 6** SICA forecast results. Table reads the same as Table 4. The results indicate IceNet's ability to reduce larger errors and high forecast skill on deviations from the climatological mean. The ordering of the GNN's compared to the SIC forecast results has also changed, with the lightweight GraphSAGE model leading for this task.

| Model | Family | MAE full | RMSE full | MAE ice | RMSE ice | Skill% (pers.) | Skill% (clim.) |
|---|---|---|---|---|---|---|---|
| Interaction | GNN | 0.0176 ± 0.0005 | 0.0566 ± 0.0004 | 0.0619 ± 0.0009 | 0.1098 ± 0.0007 | +23.3 | +14.4 |
| GraphSAGE | GNN | **0.0172 ± 0.0006** | 0.0561 ± 0.0003 | 0.0613 ± 0.0004 | 0.1090 ± 0.0006 | +25.0 | +16.3 |
| DeepGAT | GNN | 0.0183 ± 0.0011 | 0.0570 ± 0.0006 | 0.0629 ± 0.0005 | 0.1105 ± 0.0011 | +20.1 | +10.8 |
| Diffusion (DConv) | GNN | 0.0176 ± 0.0006 | 0.0565 ± 0.0001 | 0.0626 ± 0.0004 | 0.1097 ± 0.0003 | +23.2 | +14.3 |
| IceNet | CNN | 0.0176 ± 0.0002 | **0.0556 ± 0.0003** | **0.0607 ± 0.0003** | **0.1077 ± 0.0005** | +23.1 | +14.3 |
| SICNet-CBAM | CNN | 0.0202 ± 0.0011 | 0.0583 ± 0.0005 | 0.0631 ± 0.0011 | 0.1125 ± 0.0009 | +12.0 | +1.8 |
| SICNet-TSAM | CNN | 0.0212 ± 0.0009 | 0.0586 ± 0.0004 | 0.0640 ± 0.0008 | 0.1130 ± 0.0007 | +7.7 | −3.0 |
| Unicorn | CNN | 0.0242 ± 0.0016 | 0.0605 ± 0.0005 | 0.0672 ± 0.0010 | 0.1156 ± 0.0008 | −5.8 | −18.0 |
| ANTSIC-UNet | CNN | 0.0199 ± 0.0006 | 0.0580 ± 0.0008 | 0.0644 ± 0.0009 | 0.1121 ± 0.0016 | +13.0 | +2.9 |
| Persistence | Baseline | 0.0229 | 0.0742 | 0.0863 | 0.1440 | – | – |
| Climatology | Baseline | 0.0205 | 0.0715 | 0.0771 | 0.1386 | – | – |

subset. The relative performance of the learned models is consistent with that observed on the full test set. The dynamical model is outperformed by every ML model but also by both statistical baselines. Part of this will be due to the forced downscaling of the model's prediction resolution, which is particularly punishing at the ice edge. However, when we put the Interaction GNN through the same interpolation pipeline ($1 \rightarrow 1.5 \rightarrow 1^{\circ}$), the gap between the models only shrinks by about 16%. This suggests that there is a genuine skill difference and not just a systematic offset. The superior performance of ML models over dynamical models is a consistent reoccurrence within the Antarctic sea-ice literature (Sect. "Antarctic sea ice prediction").

## SICA forecast results

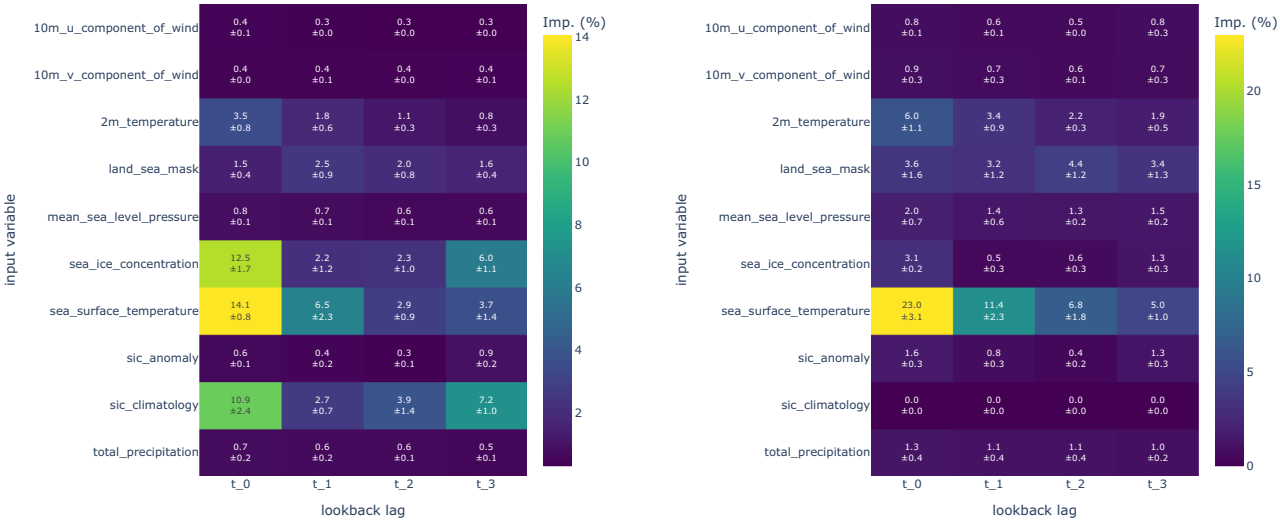
The anomaly field of SIC, as defined in Eq. 1), represents the sea ice's departure from the climatological seasonal cycle. This task is thus arguably a more direct test of genuine forecasting ability. We repeat the cross-model comparison of Table 4 on the SICA task, again over five seeds, with the same models, metrics and domains. The results of the models on this task are summarised in Table 6. Because the climatological expectation of an anomaly is zero, the climatology baseline here corresponds to a trivial predict-zero-anomaly forecast, so the skill-vs-climatology column essentially measures whether a model improves on doing nothing.

The results on the SICA forecast task show a familiar picture compared to the SIC task but with a telling difference. While the GNNs again outperform most CNN-based models on all metrics and domains, the IceNet model dominates with its unique ability to reduce large errors. IceNet posts the best full-grid RMSE ($0.0556 \pm 0.0003$), ice-masked MAE ($0.0607 \pm 0.0003$) and ice-masked RMSE ($0.1077 \pm 0.0005$). This is consistent with, and sharper than, the SIC-task pattern already noted in the previous subsection: removing the large, near-static climatological signal from the target removes exactly the part of the full-grid error where the GNN's advantage was concentrated, leaving IceNet's strength at suppressing large ice-edge errors as the dominant factor on all but the average full-grid error. Framed against the two baselines, the GNN variants and IceNet are the only models that clear climatology comfortably in double digits (up to +16.3% for GraphSAGE), while SICNet-CBAM, SICNet-TSAM and especially Unicorn barely beat, or fail to beat, the trivial predict-zero-anomaly forecast. Within the GNN family, the operator ranking also reshuffles relative to SIC: GraphSAGE is here the strongest operator and DeepGAT the weakest.

## Feature attribution

Data-driven forecasts are often treated as black boxes, since it is not obvious which inputs actually drive their predictions. To interpret what our model has learned, we applied **Integrated Gradients** (IG)[39], a gradient-based feature-attribution method, to the trained Interaction GNN. We used two reference baselines: a **zero baseline**, which measures importance relative to having no information at all, and a **climatology baseline**, which measures importance relative to the seasonal-normal state and thereby isolates the signal a model uses to beat climatology (mirroring the SICA task of Equation 1). For more details on the IG method, baselines, and averaging protocol, see Sect. "Feature-attribution method". The results are summarised in the two heatmaps in Fig. 7. They report each input feature's share of the total attribution (mean ± standard deviation over five seeds) under both baselines.

The heatmaps show that first, under the zero baseline, the model's forecast is driven predominantly by the state of the ice–ocean system rather than the atmosphere: sea-surface temperature (27.2%), the SIC climatology (24.7%) and sea-ice concentration (22.9%) together account for approximately 75% of the total attribution. The




(a) Zero baseline.

(b) Climatology baseline.

**Fig. 7** Input feature importance as interpreted by Integrated Gradients. Heatmap of the IG results on (**a**) zero and (**b**) the climatology baseline. The vertical axis lists the input variables and the horizontal the lookback lag with `t_0` = most recent. Numbers represent the share of total absolute attribution (%), mean ± standard deviation over five training seeds. (**a**) Zero baseline: sea-surface temperature, SIC climatology and sea-ice concentration co-dominate (≈75% combined). (**b**) Climatology baseline (importance relative to the seasonal-normal state): sea-surface temperature dominates (≈46%) while the SIC-climatology channel collapses to 0%, isolating SST as the primary driver of skill beyond climatology.

atmospheric inputs (mean sea-level pressure, 10 m winds, 2 m temperature and total precipitation), on the other hand, individually contribute no more than about 7%. The physical interpretation is: on the sub-seasonal (multi-week) time-scales considered here, Antarctic sea ice is governed primarily by its own recent state, the underlying ocean temperature, and the strong climatological seasonal cycle, rather than by short-term atmospheric variability. The land–sea mask adds another 7.7%, which is reasonable, as it encodes a static geographic constraint on where ice can physically exist.

Second, switching to the climatology baseline changes the attribution pattern substantially. The SIC-climatology channel drops to exactly 0%, which makes sense since it now **is** the baseline. Sea-ice concentration also drops sharply, from 22.9% down to 5.4%. **Sea-surface temperature, however, becomes even more dominant, rising to 46.2%** of the total importance as it is essentially picking up most of the weight that climatology and SIC leave behind. Land-sea geography (14.7%) and near-surface air temperature (13.5%) become the next most important inputs, both clearly bigger than under the zero baseline (7.7% and 7.3%). The other weather variables stay minor throughout (each $\leq$ 6.2%). Put together, the two baselines complement each other well: the zero baseline shows the model anchoring on the recent ice/ocean/climatological state, as expected of a skilful short-lead forecaster, while the climatology baseline reveals the genuinely predictive signal and pins it to sea-surface temperature as the primary driver of ice evolution **beyond** the seasonal cycle.

Lastly, the model does not rely solely on the most recent observation, under either baseline. While the most recent week (`t_0`) is clearly most important to the prediction, for the zero baseline, the oldest of the four input weeks (`t_3`) is the second most important at about 22%, ahead of the two weeks in between (18% and 14%). This suggests the model is not only reading the most recent snapshot, but also comparing the oldest and newest weeks to pick up on a short-term trend, rather than treating each week independently.

## Parameter and compute efficiency

Table 7 compares model size and per-sample compute to the MAE and RMSE (on the ice-edge) results. The GNNs have between 4× and 24× fewer trainable parameters than the CNN benchmarks while exceeding them on three of the four metrics (Table 4). Parameter efficiency does not, however, translate into cheap inference, as message passing over 18, 000 grid nodes and the associated edge sets costs $\approx$ 119 GFLOPs per forward pass for the Interaction GNN, which is $\approx$ 5.4× larger than the smallest, most efficient CNN (IceNet) but below Unicorn's 148 GFLOPs. The average characterisation of the GNN models compared to the CNNs is therefore that they might be *small* but not *cheap*. They compress the forecasting task into far fewer parameters, at the price of higher per-sample compute.

**Table 7** Model parameters and efficiency. Trainable parameter count, checkpoint size, and per-forward-pass compute (multiply–accumulate operations (MAc) and floating-point operations (FLOP) for a single input sample) for every learned model, alongside its predictive skill on the dynamic-ice mask. Parameter counts include trainable parameters only; fixed graph buffers such as edge-index tensors are excluded. The GNNs match or exceed the CNNs' ice-masked MAE with 4–24× fewer parameters, but require higher per-sample compute.

| Model | Family | Parameters (M) | Size (MB) | GMACs | GFLOPs | MAE ice | RMSE ice |
|---|---|---|---|---|---|---|---|
| GraphSAGE | GNN | 1.29 | 5.16 | 55.72 | 111.45 | 0.060402 | 0.112215 |
| DeepGAT | GNN | 1.29 | 5.17 | 56.39 | 112.78 | 0.060378 | 0.112040 |
| Diffusion (DConv) | GNN | 1.55 | 6.21 | 56.20 | 112.39 | 0.062151 | 0.116296 |
| Interaction | GNN | 1.95 | 7.79 | 59.57 | 119.14 | 0.060017 | 0.112066 |
| IceNet | CNN | 7.81 | 31.23 | 11.13 | 22.26 | 0.064503 | 0.107928 |
| SICNet-TSAM | CNN | 18.15 | 72.58 | 18.93 | 37.86 | 0.064830 | 0.113307 |
| SICNet-CBAM | CNN | 18.26 | 73.03 | 18.93 | 37.86 | 0.064371 | 0.113703 |
| Unicorn | CNN | 29.18 | 116.70 | 73.79 | 147.58 | 0.061753 | 0.114334 |
| ANTSIC-UNet | CNN | 31.41 | 125.64 | 15.41 | 30.81 | 0.063035 | 0.114099 |

## Discussion and future work

While this study seeks to introduce the first analysis of graph-based Antarctic SIC forecasting by conducting a comprehensive graph creation, GNN pipeline design and ablation study, it did have a few caveats worth discussing before we summarise its contributions in the conclusion.

On the graph modelling side, we did not sweep the SLIC construction hyperparameters (compactness, segment budget and pre-smoothing) or ablate the input channels, so the effect of graph geometry cannot be fully separated from these fixed choices. As each of these design choices requires a wrapper around the GNN model, tuning them requires significant training and compute. This is also why this study conducted both ablations in sequence. The operator comparison was thus only run on a single graph resolution, i.e. the best-performing graph of the previous experiment. This leaves operator–graph interactions untested. On the comparison itself, all CNN results are our own re-implementations under a common experimental setup rather than the originally published numbers, and were not individually re-tuned to the fullest extent. It is also worth highlighting that the dynamical-model comparison relies on a single deterministic forecast on a smaller matched subset that had to be downscaled onto our grid, so it is not strictly like-for-like. Finally, every seed-based comparison in this paper uses a robustness heuristic (twice the across-seed standard deviation) rather than a formal significance test. Many of the margins we report are small, and should be read with that in mind.

With these shortcomings come opportunities for future work. The most direct is to conduct a complete full-width grid search over the SLIC construction parameters. A second direction could exploit the very property that motivated a graph-based approach, which is the freedom to encode arbitrary, long-range connectivity. The graphs used here are built predominantly from spatial adjacency, whereas future work could add edges that link the ice edge to remote climate drivers, or learn the graph structure directly from data, testing whether explicit teleconnections improve skill at longer lead times. This is particularly promising given that our feature-attribution analysis identified sea-surface temperature as the dominant driver of skill beyond the seasonal climatology. Finally, the present framework is deterministic and trains a separate model per task. Extending it to probabilistic, ensemble forecasts would supply the calibrated uncertainty that operational use requires, while a fairer native-resolution comparison against an ensemble dynamical forecast, together with transferring the pipeline to the Arctic, would test how far these findings generalise.

## Conclusion

To the best of our knowledge, this is the first study to investigate whether operating on arbitrary graph topologies rather than a fixed Euclidean grid offers any advantage for forecasting Antarctic sea ice concentration (SIC) and its anomalies (SICA), a task that convolutional neural networks (CNNs) and operational dynamical models have so far dominated. To do so, we built the first graph neural network (GNN) forecasting framework for this region, evaluated two different ways of turning the gridded observations into a graph (an icosahedral multi-mesh and a data-adaptive SLIC super-pixel graph), ablated both the graph construction and the message-passing operator, compared the resulting model against five re-implemented CNN benchmarks, two statistical baselines and the operational ECMWF forecast, and used feature attribution to probe what the model has learned.

The ablation showed that for our pipeline, how the coarse graph is constructed matters more than which message-passing operator runs on it. The hierarchical connectivity of the icosahedral multi-mesh and of the higher-level SLIC graphs is the most influential property to boost performance. The actual difference between the largest SLIC and best-performing mesh is still within seed noise, making both graphs practical choices. The choice of the processor within the encoder–processor–decoder architecture, on the other hand, seems much less significant in comparison. The three main operators (Interaction, GraphSAGE, DeepGAT) are statistically indistinguishable, with only the diffusion operator clearly weaker and less stable to train. This suggests that, for this problem, the

performance advantages stem from the spatial scaffold of the graph that carries information across the domain rather than in the specific aggregation rule.

The final evaluation of the best-performing graph and model from the ablation study showed that the GNNs are the stronger models on average error compared to the baselines, dynamical model and CNN benchmarks. They lead on mean absolute error (MAE) on both the full grid and the ice edge, with a large margin and achieve this with far fewer parameters (roughly 4–24× smaller than the CNN counterparts). When evaluating large instead of average errors using the RMSE metric, the picture changes slightly. While the GNNs still perform well on the full grid, the U-Net-based IceNet model takes the lead with a 4% margin on the dynamic ice mask. A spatial difference analysis showed that this error lead is concentrated mainly at the inner ice ridge but is consistent across the regions. Notably, every machine-learning model outperforms the operational dynamical forecast, reinforcing a pattern already reported in the Antarctic sea-ice literature.

Lastly, the feature attribution analysis revealed that sea-surface temperature is the dominant driver of forecast skill beyond the seasonal climatology, which is physically sensible. However, this only identified what the GNN model attends to, not what causally drives sea-ice change.

## Methods

This section provides the full data, graph, model, and evaluation details required to reproduce the study, supplementing the condensed descriptions in the main text.

### Graph construction

The GNN operates on a fine-resolution spatial grid coupled to a coarser graph that provides long-range spatial connectivity. Let $G_g = (V_g, E_g)$ denote the $50 \times 360$ input grid, where $|V_g| = N_g = 18{,}000$ and each grid node $i \in V_g$ carries an input feature vector $\mathrm{x}_i \in \mathbb{R}^F$, with $F = 40$. The coarse graph is denoted by $G_c = (V_c, E_c)$. The complete graph used by the encoder–processor–decoder architecture is therefore

$$G = \left(V_g \cup V_c,\, E_g \cup E_{\text{enc}} \cup E_c \cup E_{\text{dec}}\right), \qquad E_{\text{dec}} = \left\{(v_c, v_g) \mid (v_g, v_c) \in E_{\text{enc}}\right\}. \tag{2}$$

The two graph constructions differ only in the definition of the coarse node set $V_c$ and the coarse edge set $E_c$.

#### Icosahedral multi-mesh

The first graph is a geometry-based hierarchical icosahedral mesh. Starting from an icosahedron with 12 vertices and 30 edges, each refinement level subdivides every triangular face into four smaller faces. After $K$ refinement levels, the mesh contains

$$|V_{\text{mesh}}^{(K)}| = 10 \cdot 4^K + 2 \tag{3}$$

vertices. Rather than replacing edges at each refinement level, edges from all levels are retained. Thus, with $E_{\text{new}}^{(k)}$ denoting the edges introduced at level $k$, the multi-scale mesh is

$$G_{\text{mesh}}^{(K)} = \left(V_{\text{mesh}}^{(K)}, E_{\text{mesh}}^{(K)}\right), \qquad E_{\text{mesh}}^{(K)} = \bigcup_{k=0}^{K} E_{\text{new}}^{(k)}. \tag{4}$$

Each grid node is connected to its three nearest mesh nodes in three-dimensional Cartesian space. The encoder and decoder relations are therefore

**Table 8** Multi-mesh graph properties. "Edges at level $K$" denotes the edges added at that refinement within the AOI, whereas "cumulative edges" indicates the total number of edges that have been included up to level $K$.

| Level $K$ | Nodes | Edges at level $K$ | Cumulative edges |
|---|---|---|---|
| 1 | 6 | 10 | 10 |
| 2 | 31 | 75 | 85 |
| 3 | 106 | 285 | 370 |
| 4 | 441 | 1,255 | 1,625 |
| 5 | 1,801 | 5,265 | 6,890 |
| 6 | 7,206 | 21,340 | 28,230 |
| Encoder/decoder edges: 3x18,000 = 54,000 | | | |
| Avg. grid→mesh in-degree: 30.0 ($K$=5), 7.5 ($K$=6) | | | |

**Table 9** SLIC super-pixel graph properties. The table lists the edge counts by level, at two segment budgets (n=1,800) and (n=7,200). Node numbers are constant, as each level only adds new edges to existing nodes within the new clusters and "promotes" starter nodes to super-nodes. Rows run coarsest (top) to finest (bottom); level 0 represents the finest region segmentation with only one level of super-node edges, while higher levels are coarser hierarchical "highway" edges.

| | SLIC (**n = 1800**) | SLIC (**n = 7200**) |
|---|---|---|
| Nodes | 1,805 | 4,486 |
| Avg. grid in-degree | 10.0 | 4.0 |
| Level $L$ | | |
| 5 | – | 12 |
| 4 | 12 | 61 |
| 3 | 61 | 257 |
| 2 | 256 | 1,036 |
| 1 | 1,007 | 4,866 |
| 0 | 4,962 | 11,863 |
| Total graph edges | 6,298 | 18,095 |
| Encoder/decoder edges: 18,000 | | |

$$E_{\text{enc}}^{\text{mesh}} = \left\{(i,j) \in V_g \times V_{\text{mesh}}^{(K)} \mid j \in 3\text{NN}(i)\right\}, \qquad E_{\text{dec}}^{\text{mesh}} = \left\{(j,i) \mid (i,j) \in E_{\text{enc}}^{\text{mesh}}\right\}. \tag{5}$$

Table 8 summarises the mesh graph properties based on the refinement level $K$.

### SLIC super-pixel graph

The SLIC algorithm is performed on z-scored data from the training-period temporal mean, so that construction is leakage-free with respect to the validation and test data. We use the scikit-image implementation[40] with a target segment count $n$, compactness 0.1, and pre-smoothing $\sigma = 1.0$; the low compactness makes region boundaries follow the climatological ice structure rather than a regular tiling. We furthermore augmented the edge creation by a wrap-around seam (column $0 \leftrightarrow 359$ in each grid row) so that the circumpolar ring closes.

Long-range connectivity is added hierarchically. Coarser levels are produced by re-segmenting the same image at successively smaller budgets (dividing $n$ by four per level, mirroring the mesh's four-fold growth); for each coarse region, a representative is chosen from the finest-level nodes by dominant overlap, and the coarse adjacency is added as level-tagged "highway" edges between representatives. The node set is unchanged throughout; coarser levels add only edges in direct analogy to the union-of-levels multi-mesh. Table 9 reports the per-level edge counts for the two budgets used in the ablation: $n = 1{,}800$ (1,805 nodes) and $n = 7{,}200$ (4,486 realised nodes). The formal notation of the graph: Let

$$\mathrm{Z} \in \mathbb{R}^{H \times W \times C}, \qquad (H, W, C) = (50, 360, 3), \tag{6}$$

denote the climatological input used for segmentation. At hierarchy level $l$, SLIC partitions the grid into $n_l$ non-overlapping regions,

$$\mathcal{P}^{(l)} = \left\{R_1^{(l)}, \ldots, R_{n_l}^{(l)}\right\}, \qquad \bigcup_{a=1}^{n_l} R_a^{(l)} = V_g, \qquad R_a^{(l)} \cap R_b^{(l)} = \emptyset \quad (a \neq b). \tag{7}$$

Each region is represented by a super-node $v_a^{(l)}$, located at the centroid

$$\mathrm{p}_a^{(l)} = \frac{1}{|R_a^{(l)}|} \sum_{i \in R_a^{(l)}} \mathrm{p}_i, \qquad V_{\mathrm{SLIC}}^{(l)} = \left\{v_1^{(l)}, \ldots, v_{n_l}^{(l)}\right\}. \tag{8}$$

The complete hierarchical SLIC graph is

$$G_{\mathrm{SLIC}} = (V_{\mathrm{SLIC}}, E_{\mathrm{SLIC}}), \qquad V_{\mathrm{SLIC}} = \bigcup_{l=1}^{L} V_{\mathrm{SLIC}}^{(l)}, \tag{9}$$

$$E_{\mathrm{SLIC}} = \left(\bigcup_{l=1}^{L} E_{\mathrm{SLIC}}^{(l)}\right) \cup \left(\bigcup_{l=1}^{L-1} E_{\mathrm{inter}}^{(l,l+1)}\right), \tag{10}$$

where $E_{\mathrm{SLIC}}^{(l)}$ connects spatially adjacent super-nodes within level $l$, while $E_{\mathrm{inter}}^{(l,l+1)}$ connects corresponding regions between adjacent hierarchy levels.

The grid-to-graph mapping follows the segmentation:

$$E_{\mathrm{enc}}^{\mathrm{SLIC}} = \left\{(i, v_a^{(l)}) \mid i \in R_a^{(l)}\right\}, \qquad E_{\mathrm{dec}}^{\mathrm{SLIC}} = \left\{(v_a^{(l)}, i) \mid (i, v_a^{(l)}) \in E_{\mathrm{enc}}^{\mathrm{SLIC}}\right\}. \tag{11}$$

## GNN architecture

Two additional design details are worth noting about the encoder-processor-decoder architecture. First, the coarse nodes carry no input data of their own as each starts from a small learnable initial vector (initialised at ∼0.01 scale), so that on the first pass their state is dominated by what the encoder delivers from the grid. Second, for the multi-mesh and hierarchical SLIC graph, each edge's refinement level is mapped to a learned edge embedding that the processor can read, allowing it to treat long- and short-range edges differently.

### Message passing: interaction network layers

The default building block for every phase is the Interaction Network layer[15], a general form of learned message passing. In one layer, each node $i$ is updated in three steps:

1. **Message:** for every edge $j \to i$, a message $m_{ij} = \mathrm{MLP}_{\mathrm{edge}}([h_i, h_j, e_{ij}])$ is computed as a learned nonlinear function of **both** endpoint states $h_i, h_j$ and, where available, an edge attribute $e_{ij}$ (in the processor, the refinement-level embedding);
2. **Aggregation:** messages arriving at node $i$ are summed, $m_i = \sum_j m_{ij}$;
3. **Update:** $h_i \leftarrow \mathrm{LayerNorm}(h_i + \mathrm{MLP}_{\mathrm{node}}([h_i, m_i]))$, where the residual connection and layer normalisation keep the eight-layer processor stack stable.

The defining property of the "interaction" is that each message depends jointly and nonlinearly on sender and receiver, which makes this layer more expressive than aggregation schemes that average neighbour features or weight them with attention before a linear transformation. The encoder and decoder use a bipartite variant of the same layer, in which messages flow from one node set to the other (grid → graph or graph → grid) and only the destination nodes are updated; the grid–coarse edges carry no edge attribute.

### Graph sample & aggregate network (GraphSAGE)

The GraphSAGE model[16] is an inductive framework designed to generate node embeddings for data not encountered during training by utilising node features. Unlike traditional approaches that require the entire graph during training, GraphSAGE employs an inductive strategy, allowing it to generate embeddings for new, unseen nodes without requiring the model to be retrained. The key idea behind GraphSAGE is to sample a fixed number of neighbouring nodes for each target node and then aggregate their features using simple functions, such as mean or max pooling. This allows the model to efficiently capture the local structure of the graph without needing to process every neighbour, which would be computationally expensive in very large graphs. This approach makes GraphSAGE well-suited for real-world applications such as recommendation systems, where user behaviour and relationships are constantly evolving. However, it has recently also shown promising results in the environmental domain ([12,41]). It is edge-unaware (the refinement-level embedding is ignored) and serves as a simpler baseline in which the aggregation is fixed rather than learned per edge.

### DeepGAT

Graph attention networks (GAT)[17] represent a modification of graph neural architectures through the use of dynamic attention mechanisms. While standard GNNs, such as graph convolutional networks (GCN), employ static aggregation schemes, GATs learn adaptive attention coefficients that capture varying influence patterns between connected nodes. This mechanism can help learn not just spatial, but also temporal dependencies. As a result, attention-based models have famously become popular alternatives to RNN architectures as they reduce the computational demands of classical RNNs. It is edge-aware and learns how strongly to weight each neighbour, without constructing a nonlinear joint function of sender and receiver as the Interaction layer does. In our experiments, the DeepGAT uses four attention heads whose outputs are concatenated (each head of width $128/4 = 32$, giving 128 channels in total) and without added self-loops; the refinement-level embedding is passed as the edge attribute.

### Diffusion convolution (DConv)

The final operator is the diffusion convolution (DConv). Diffusion convolution frames message passing as a $K$-step diffusion process on the graph. Node features are propagated along powers of the (degree-normalised) transition matrix, so each node aggregates from its $K$-hop neighbourhood weighted by random-walk transition probabilities[18]. We adopt the bidirectional random-walk formulation of the Diffusion Convolutional Recurrent Neural Network (DCRNN) by Li et al.[19], which diffuses along both the forward and reverse transition matrices so that directed structure is respected. However, we do not use the diffusion-convolutional gated recurrent unit (DCGRU) that defines the full DCRNN model. Instead, temporal dependencies in our architecture are handled by the separate GRU rollout decoder, which is shared across all operator variants. Like GraphSAGE, this operator is edge-unaware and thus uses only the (unit-weighted) graph structure and cannot exploit the per-edge refinement-level embedding. The diffusion-convolution operator in our study is taken directly from the PyTorch Geometric Temporal library[42], with $K=2$ bidirectional random-walk diffusion steps. Because the operator normalises by node degree, self-loops are added to avoid division by zero at isolated coarse nodes, and duplicate edges are removed. The implementation materialises a dense adjacency matrix over the batched coarse nodes at every forward pass, which makes it substantially more memory-intensive than the other operators and caps the usable batch size (2 in our setting).

## Graph ablation

The two constructions differ in three structurally important ways, held constant across all configurations. First, the mesh budgets are dictated by the mesh's geometrically fixed refinement steps which results in $K = 5$ being a roughly 10-fold coarsening (the lowest reduction we considered), and $K = 6$ being the largest usable resolution ($\approx$ 2.5-fold reduction) before $K = 7$ (28,766 nodes) would exceed the grid itself. The SLIC budgets were then set to match the mesh node counts as closely as possible. At matched budgets they have comparable overall connectivity with the $K = 5$ mesh and the hierarchical $n = 1{,}800$ SLIC graph carrying similar total edge counts (Tables 8 and 9; $\approx$9% fewer for SLIC) and mean node degree (7.7 versus 7.0), and both reach an emergent finest-level degree of $\approx 4$ ($\approx$3.8–3.9 for the mesh). Second, they attach the 18,000-cell grid differently: the mesh's three-nearest-neighbour rule pools each grid cell into three *overlapping* mesh nodes ($\approx$30 cells per node at $K = 5$, $\approx$7.5 at $K = 6$), whereas SLIC is a hard partition into *disjoint* regions ($\approx$10 cells per region at $n = 1{,}800$, $\approx$4 at $n = 7{,}200$ ), so SLIC's grid-to-graph fan-in is one-third of the mesh's. Third, the mesh refines essentially without bound ($\times$4.0 nodes and $\times$4.1 edges per level), whereas SLIC saturates near 4,500 regions on this grid, therefore, the high-resolution SLIC graph has 38% fewer nodes and 36% fewer edges than the $K = 6$ mesh (a $\times$2.5 node / $\times$2.9 edge increase from low to high resolution, versus $\times$4 for the mesh).

## Benchmarks

### Dynamical model (ECMWF)

In addition to the baselines, we compare the results with forecasts from ECMWF's numerical weather prediction model. The ECMWF S2S forecasting system features an interactive sea ice component, utilising the Louvain-la-Neuve Sea Ice Model (LIM2), which dynamically simulates the evolution of sea ice in response to atmospheric–ocean interactions. Uncertainty is represented by perturbing the initial sea ice state using five ensemble members from the ocean–sea ice reanalysis, while no stochastic perturbations are applied during the forecast. As a result, the ECMWF sea ice predictions reflect physically consistent dynamical evolution, though they remain sensitive to biases in the coupled model and to errors in the initial reanalysis. The CY43R1 version used in this study provides predictions since November 2016, twice a week for up to 46 days in advance. Because of this limited number of prediction data, we created a subset of the full test set which is adjusted to the prediction windows and period of the ECMWF model. This subset consists of 498 initialisation dates at forecast leads of 1–6 weeks (matching the 46-day prediction horizon). Furthermore, the sea ice variables are available at 1.5° spatial resolution, so we used linear interpolation to downscale the resolution to 1° to match our dataset.

### CNN benchmarks

We re-implement five published convolutional architectures for sea-ice prediction and train them from scratch under the protocol above; each receives the same 40 input channels (10 predictors $\times$ 4 lookback weeks) and emits the same 8-week forecast. We emphasise that all CNN results in this paper are therefore **our re-implementations under a common experimental setup**, not the numbers reported in the original publications, which target different regions, resolutions, horizons, and task formulations (see Sects. "Arctic sea ice prediction" and "Antarctic sea ice prediction"). Where the original designs were incompatible with our regression task, we made minimal adaptations:

IceNet[20] was originally a 25-member probabilistic ensemble predicting three SIC classes over a 6-month horizon. We adapt it to deterministic regression with a single convolutional output head, sigmoid activation, and MSE loss over the 8-week horizon.

SICNet-CBAM and SICNet-TSAM[21] share a five-stage encoder backbone; the two variants differ in their attention module (convolutional block attention versus a temporal self-attention module built from causal dilated convolutions). We train both as published, adapted only to our input/output specification.
Unicorn[22] combines a continuous neural ODE (ConvNODE) backbone with a decomposed multi-period module that extracts trend components along the temporal axis. We substitute GroupNorm for BatchNorm inside the ODE function for integration stability and cap the decomposition kernel at the 4-week lookback.
ANTSIC-UNet[23] is a U-Net encoder–decoder for Antarctic SIC. We replace the original stacked-sigmoid output head with parallel per-week sigmoid heads, which avoids vanishing gradients from stacked sigmoid activations across the forecast weeks.

## Feature-attribution method

Data-driven ML forecasts are often treated as black boxes, as it is not obvious which inputs actually drive the model's predictions. To help interpret what our model has learned, we used **Integrated Gradients** (IG)[39], a gradient-based feature-attribution method. IG starts from a "blank" reference input and gradually morphs it into the real input while tracking how much each input channel pushes the prediction towards the right direction. This yields, for every input channel, a signed contribution that sums (approximately) to the difference between the baseline and the actual prediction. It is a way of attributing a model's output back to its inputs. We applied this to the five seeds of the trained Interaction GNN across 10 test windows with 16 path-integral steps and averaged the results. We report each channel's importance as its % share of the total attribution, together with how much this share varies across the five seeds (mean ± standard deviation).

What counts as "blank" matters, so we tried two different reference points. The **zero baseline** compares the real input to an input of all zeros, which asks "how important is this input compared to having no information at all?" Because SIC and its climatology are always large relative to zero, however, this baseline risks crediting the model for simply having sensible inputs rather than for extracting forecast-relevant information from them. We therefore also computed a **climatology baseline**, in which the SIC channel is set to that week's climatological value, the SIC anomaly channel to zero, and all z-scored dynamic channels to their training-period mean. This asks the sharper question: "once the model already knows the normal seasonal state, what extra information is it using to do better than that?" This mirrors the SICA (anomaly) task defined earlier (Equation 1), which is built around the same idea.

## Training and hyperparameter optimization

The model is trained with a mean-squared-error loss over all eight forecast weeks using the AdamW optimiser, with early stopping on the validation loss. The key optimisation hyperparameters were selected beforehand by a Bayesian search using Optuna[43] (a Tree-structured Parzen Estimator sampler with median pruning), we ran a 30-trial study over the learning rate (log-uniform $10^{-4}$–$3 \times 10^{-3}$), the AdamW weight decay (log-uniform $10^{-6}$–$10^{-2}$), and the learning-rate schedule (ReduceLROnPlateau vs. linear-warmup cosine, with their respective sub-parameters), minimising the validation loss. The resulting configuration (learning rate $\approx 6.5 \times 10^{-4}$, weight decay $\approx 7.3 \times 10^{-4}$, plateau schedule) was then held fixed and reused across all models, tasks, and seeds, so that the architectural comparisons in Sect. "Ablation results" are not confounded by per-model tuning. All models were trained on NVIDIA A100 (80 GB) GPUs on the NeSI Mahuika cluster, scheduled via SLURM.

Non-diffusion models used a batch size of 16 graphs; the diffusion operator was capped at a batch size of 2 owing to its dense $(B \cdot n_{\text{mesh}})^2$ adjacency. Training ran for at most 300 epochs with early stopping (patience of 30 epochs without validation-loss improvement), restoring the best checkpoint. The selected ReduceLROnPlateau schedule multiplied the learning rate by 0.5 after 7 epochs without validation-loss improvement (relative threshold $10^{-4}$), with no lower bound on the learning rate (minimum 0).

**Table 10** Notation used in the graph formulation.

| Symbol | Dimension / type | Description |
|---|---|---|
| $G_g$ | Graph | Fine-resolution spatial grid graph |
| $V_g$ | Set | Set of fine-resolution grid nodes |
| $N_g$ | Scalar | Number of fine-resolution grid nodes ($N_g = 18{,}000$) |
| $E_g$ | Set | Local grid-connectivity edges |
| $G_c$ | Graph | Coarse graph used for long-range message passing |
| $V_c$ | Set | Set of coarse graph nodes |
| $N_c$ | Scalar | Number of coarse graph nodes |
| $E_c$ | Set | Edges connecting coarse graph nodes |
| $G$ | Graph | Full graph combining grid and coarse graph nodes |
| $E_{\text{enc}}$ | Set | Encoder edges from grid nodes to coarse nodes |
| $E_{\text{dec}}$ | Set | Decoder edges from coarse nodes to grid nodes |
| $\mathrm{x}_i$ | $\mathbb{R}^F$ | Input feature vector at grid node $i$ |
| $F$ | Scalar | Number of input features per grid node ($F = 40$) |
| $\mathrm{h}_i$ | $\mathbb{R}^D$ | Hidden representation of node $i$ |
| $K$ | Scalar | Number of icosahedral mesh refinement levels |
| $V_{\text{mesh}}^{(K)}$ | Set | Nodes of the $K$-level icosahedral multi-mesh |
| $E_{\text{mesh}}^{(K)}$ | Set | Multi-scale edges of the icosahedral mesh |
| $E_{\text{new}}^{(k)}$ | Set | Edges introduced at mesh refinement level $k$ |
| $\ell(e)$ | $\{0, \ldots, K\}$ | Refinement level at which edge $e$ was introduced |
| $\mathrm{3NN}(i)$ | Set | Three nearest mesh nodes associated with grid node $i$ |
| Z | $\mathbb{R}^{H\times W\times C}$ | Climatological input used for SLIC segmentation |
| $\mathcal{P}^{(l)}$ | Partition | SLIC segmentation at hierarchy level $l$ |
| $R_a^{(l)}$ | Set | Grid cells belonging to SLIC region $a$ at level $l$ |
| $n_l$ | Scalar | Requested number of SLIC regions at level $l$ |
| $v_a^{(l)}$ | Node | Super-node representing SLIC region $R_a^{(l)}$ |
| $\mathrm{p}_a^{(l)}$ | $\mathbb{R}^2$ | Spatial centroid of SLIC region $R_a^{(l)}$ |
| $E_{\text{SLIC}}^{(l)}$ | Set | Within-level edges between adjacent SLIC regions |
| $E_{\text{inter}}^{(l,l+1)}$ | Set | Inter-level edges between adjacent SLIC hierarchy levels |
| $SIC_w$ | Spatial field | Weekly sea-ice concentration at forecast week $w$ |
| $SICA_w$ | Spatial field | Weekly sea-ice concentration anomaly at forecast week $w$ |
| $SIC_w^{\text{clim}}$ | Spatial field | Climatological SIC for forecast week $w$ |
| $w$ | Scalar | Forecast-week index |

### Software environment

All experiments used Python 3.11.15 with PyTorch 2.13.0 (CUDA 13.0) and PyTorch Geometric 2.8. Graph and super-pixel construction additionally used scikit-image 0.26.0 and SciPy 1.17.1. The diffusion operator was trained in a separate environment with PyTorch 2.5.1 (CUDA 12.1) with PyTorch Geometric Temporal 0.54.0, because `torch-geometric-temporal` (and its `torch-scatter`/`torch-sparse` dependencies) provides no prebuilt wheels compatible with the main environment's PyTorch version; all other models and operators share a single environment.

## Appendix

See Table 10.

**Author contributions** T.M. wrote the main manuscript text and prepared the figures. All authors reviewed the manuscript

**Funding** This work was supported by the New Zealand Ministry of Business, Innovation, and Employment (MBIE) through the Time-Evolving Data Science/Artificial Intelligence for Advanced Open Environmental Science project (TAIAO)

[CONT-64517-SSIFDS-UOW] (URL: https://taiao.ai/) and the Antarctic Science Platform [ANTARCTICANZ2504].

**Data availability** The preprocessed datasets as described in this work are available via this URL: https://www.kaggle.com/datasets/tobiasmilz/sicgraph/data. For the code or help with loading the data, please contact the authors via the data repository.

**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

## References

1. Arrigo, K.R. Sea Ice Ecosystems. *Annual Review of Marine Science*, 6 (Volume 6, 2014):439–467, 1 (2014). ISSN 19411405. https://doi.org/10.1146/ANNUREV-MARINE-010213-135103/CITE/REFWORKS. https://www.annualreviews.org/content/journals/10.1146/annurev-marine-010213-135103.
2. Rysgaard, S., Bendtsen, J., Delille, B., Dieckmann, G.S., Glud, R.N., Kennedy, H., Mortensen, J., Papadimitriou, S., Thomas, D.N. & Tison, J.L. Sea ice contribution to the air–sea CO2 exchange in the Arctic and Southern Oceans. *Tellus B: Chem. Phys. Meteorol.*, **63** (5) 823–830, (2011). ISSN 02806509. https://doi.org/10.1111/J.1600-0889.2011.00571.X. https://www.tandfonline.com/doi/abs/10.1111/j.1600-0889.2011.00571.x.
3. Meier, W.N., Perovich, D., Farrell, S., Haas, C., Hendricks, S., Petty, A.A., Webster, M., Divine, D., Gerland, S., Kaleschke, L., Ricker, R., Steer, A., Tian-Kunze, X., Tschudi, M. & Wood, K. Sea Ice. *NOAA Technical Report OAR ARC*, (2021). https://doi.org/10.25923/Y2WD-FN85. https://repository.library.noaa.gov/view/noaa/34474.
4. Goosse, H., Kay, J.E., Armour, K.C., Bodas-Salcedo, A., Chepfer, H., Docquier, D., Jonko, A., Kushner, P.J., Lecomte, O., Massonnet, F., Park, H.S., Pithan, F., Svensson, G. & Vancoppenollel, M. Quantifying climate feedbacks in polar regions. *Nat. Commun.*, **9**(1), 12 (2018). https://doi.org/10.1038/S41467-018-04173-0. https://pubmed.ncbi.nlm.nih.gov/29765038/.
5. Perovich, D.K., Jones, K.F., Light, B., Eicken, H., Markus, T., Stroeve, J. & Lindsay, R. Solar partitioning in a changing Arctic sea-ice cover. *Annals of Glaciology*, **52**(57) 192–196, (2011). https://doi.org/10.3189/172756411795931543. https://www.cambridge.org/core/journals/annals-of-glaciology/article/solar-partitioning-in-a-changing-arctic-seaice-cover/363DD9F8A62A5D542B58B4C6D543E45D.
6. Meier, W.N., Hovelsrud, G.K., Oort, B.E.H.v., Key, J.R., Kovacs, K.M., Michel, C., Haas, C., Granskog, M.A., Gerland, S., Perovich, D.K., Makshtas, A. & Reist, J.D. Arctic sea ice in transformation: A review of recent observed changes and impacts on biology and human activity. *Rev. Geophys.*, **52**(3) 185–217 (2014). https://doi.org/10.1002/2013RG000431. https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2013RG000431.
7. Turner, J., Bracegirdle, T.J., Phillips, T., Marshall, G.J. & Hosking, J.S. An Initial Assessment of antarctic sea ice extent in the CMIP5 models. *J. Climate*, **26**(5) 1473–1484 (2013). https://doi.org/10.1175/JCLI-D-12-00068.1.
8. Eayrs, C., Li, X., Raphael, M.N. & Holland, D.M. Rapid decline in Antarctic sea ice in recent years hints at future change. *Nature Geoscience*, **14**(7) 460–464 (2021) https://doi.org/10.1038/s41561-021-00768-3.
9. Liu, J., Zhu, Z. & Chen, D. Lowest antarctic sea ice record broken for the second year in a row. *Ocean-Land-Atmosph. Res.*, 2:0007, (2023). https://doi.org/10.34133/olar.0007. https://spj.science.org/doi/abs/10.34133/olar.0007.
10. Maksym, T. Arctic and antarctic sea ice change: Contrasts, commonalities, and causes. *Annual Review of Marine Science*, 11(Volume 11, 2019) 187–213, (2019) https://doi.org/10.1146/annurev-marine-010816-060610. https://www.annualreviews.org/content/journals/10.1146/annurev-marine-010816-060610.
11. Lam, R., Sanchez-Gonzalez, A., Willson, M., Wirnsberger, P., Fortunato, M., Alet, F., Ravuri, S., Ewalds, T., Eaton-Rosen, Z., Hu, W., Merose, A., Hoyer, S., Holland, G., Vinyals, O., Stott, J., Pritzel, A., Mohamed, S. & Battaglia, P. Learning skillful medium-range global weather forecasting, (2023). https://www.science.org/doi/abs/10.1126/science.adi2336.
12. Ning, Ding, Vetrova, Varvara & Bryan, KarinR. Graph-based deep learning for sea surface temperature forecasts (2023)
13. Rühling Cachay, S., Erickson, E., Bucker, A.F.C., Pokropek, E.J., Potosnak, W., Osei, S. & Lütjens, B. Graph neural networks for improved el niño forecasting. In *NeurIPS 2020 Workshop on Tackling Climate Change with Machine Learning* (2020). https://www.climatechange.ai/papers/neurips2020/86.

14. DiGirolamo, N., Parkinson, C., Cavalieri, D., Gloersen, P. & Zwally, H. Sea ice concentrations from nimbus-7 smmr and dmsp ssm/i-ssmis passive microwave data, version 2, (2022). http://nsidc.org/data/NSIDC-0051/versions/2.
15. Battaglia, P., Pascanu, R., Lai, M., Rezende, D.J. & kavukcuoglu, K. Interaction networks for learning about objects, relations and physics. In *Proceedings of the 30th International Conference on Neural Information Processing Systems*, NIPS'16, 4509–4517, Red Hook, NY, USA, (2016). Curran Associates Inc. ISBN 9781510838819.
16. Hamilton, W.L., Ying, R., Leskovec, J. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, **30**, (2017). https://proceedings.neurips.cc/paper_files/paper/2017/file/5dd9db5e033da9c6fb5ba83c7a7ebea9-Paper.pdf.
17. Velickovic, P., Cucurull, G., Casanova, A., Romero, A., Liò, P. & Bengio, Y. Graph attention networks. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, (2018), Conference Track Proceedings*. OpenReview.net, 2018. https://openreview.net/forum?id=rJXMpikCZ.
18. Atwood, J. & Towsley, D. Diffusion-convolutional neural networks, (2016). https://arxiv.org/abs/1511.02136.
19. Li, Y., Yu, R., Shahabi, C. & Liu, Y. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. In *International Conference on Learning Representations*, (2018). https://openreview.net/forum?id=SJiHXGWAZ.
20. Andersson, T. R., Hosking, J.S., Pérez-Ortiz, M., Paige, B., Elliott, A., Russell, C., Law, S., Jones, D.C., Wilkinson, J., Phillips, T., Byrne, J., Tietsche, S., Sarojini, B.B., Blanchard-Wrigglesworth, E., Aksenov, Y., Downie, R. & Shuckburgh, E. Seasonal Arctic sea ice forecasting with probabilistic deep learning. *Nature Communications*, **12**(1) 5124, 8 (2021). ISSN 2041-1723. https://doi.org/10.1038/s41467-021-25257-4.
21. Ren, Y., Li, X. & Zhang, W. A Data-Driven Deep Learning Model for Weekly Sea Ice Concentration Prediction of the Pan-Arctic During the Melting Season. *IEEE Trans. Geosci. Remote Sens.*, **60** 1–19, (2022) https://doi.org/10.1109/TGRS.2022.3177600.
22. Park, J. et al. Unicorn: U-net for sea ice forecasting with convolutional neural ordinary differential equations. *Sci. Reports***15**, 36330. https://doi.org/10.1038/s41598-025-20097-4 (2025).
23. Yang, Z. et al. Extended seasonal prediction of antarctic sea ice concentration using antsic-unet. *Cryosphere* **19**(12), 6381–6402. https://doi.org/10.5194/tc-19-6381-2025. (2025) https://tc.copernicus.org/articles/19/6381/2025/.
24. Chi, J. & Kim, H.-c. Prediction of Arctic Sea Ice Concentration Using a Fully Data Driven Deep Neural Network. *Remote Sensing*, **9**(12) 1305, (2017). https://doi.org/10.3390/rs9121305.
25. Kim, Y.J., Kim, H.-C., Han, D., Lee, S. & Im, J. Prediction of monthly Arctic sea ice concentrations using satellite and reanalysis data based on convolutional neural networks. *Cryosphere*, **14**(3) 1083–1104 (2020). https://doi.org/10.5194/tc-14-1083-2020.
26. Fritzner, S., Graversen, R. & Christensen, K. H. Assessment of high-resolution dynamical and machine learning models for prediction of sea ice concentration in a regional application. *J. Geophys. Res.: Oceans* **125**(11), 11. https://doi.org/10.1029/2020JC016277 (2020).
27. Liu, Y., Bogaardt, L., Attema, J. & Hazeleger, W. Extended Range Arctic Sea Ice Forecast with Convolutional Long-Short Term Memory Networks. *Monthly Weather Rev.*, 3 (2021). ISSN 0027-0644. https://doi.org/10.1175/MWR-D-20-0113.1.
28. Ren, Y. & Li, X. Predicting the daily sea ice concentration on a subseasonal scale of the pan-arctic during the melting season by a deep learning model. *IEEE Trans. Geosci. Remote Sens.*, **61** 1–15, (2023) https://doi.org/10.1109/TGRS.2023.3279089.
29. Lin, Y., Yang, Q., Li, X., Yang, C.-Y., Wang, Y., Wang, J., Liu, J., Chen, S. & Liu, J. Optimization of the k-nearest-neighbors model for summer Arctic Sea ice prediction. *Frontiers in Marine Science*, 10, (2023). ISSN 2296-7745. https://doi.org/10.3389/fmars.2023.1260047. https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2023.1260047.
30. Wang, Y., Yuan, X., Ren, Y., Bushuk, M., Shu, Q., Li, C. & Li, X. Subseasonal prediction of regional antarctic sea ice by a deep learning model. *Geophys. Res. Lett.*, 50(17) e2023GL104347, (2023). https://doi.org/10.1029/2023GL104347. https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2023GL104347.
31. Dong, X., Yang, Q., Nie, Y., Zampieri, L., Wang, J., Liu, J. & Chen, D. Antarctic sea ice prediction with A convolutional long short-term memory network. *Ocean Modelling*, **190** 102386 (2024). https://doi.org/10.1016/J.OCEMOD.2024.102386.
32. Lin, Y., Yang, Q., Li, X., Dong, X., Luo, H., Nie, Y., Wang, J., Wang, Y. & Min, C. Ice-kNN-South: A Lightweight Machine Learning Model for Antarctic Sea Ice Prediction. *J. Geophys. Res.: Mach. Learn. Comput.*, **2**(1):e2024JH000433, (2025). https://doi.org/10.1029/2024JH000433.
33. Baek, G., Ko, J., Jin, E.K. & Kug, J-S. Seasonal predictability of antarctic sea ice based on a deep-learning approach. *EGUsphere*, (2026). https://doi.org/10.5194/egusphere-2026-1556. preprint.
34. Price, I., Sanchez-Gonzalez, A., Alet, F., Andersson, T.R., El-Kadi, A., Masters, D., Ewalds, T., Stott, J., Mohamed, S., Battaglia, P., Lam, R. & Willson, M. Probabilistic weather forecasting with machine learning. *Nature*, **637**(8044) 84–90 https://doi.org/10.1038/s41586-024-08252-9.
35. Bailie, T., Mukkavilli, S., Vetrova, V. & Koh, Y.S. Hierarchical graph networks for accurate weather forecasting via lightweight training. (2025). https://doi.org/10.48550/arXiv.2510.22094.

36. Li, P., Yu, Y., Huang, D., Wang, Z.-H. & Sharma, A. Regional Heatwave Prediction Using Graph Neural Network and Weather Station Data. *Geophysical Research Letters*, **50**(7) e2023GL103405, 4 (2023). https://doi.org/10.1029/2023GL103405. https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2023GL103405.
37. Achanta, Radhakrishna et al. Slic superpixels compared to state-of-the-art superpixel methods. *IEEE Trans. Pattern Anal. Mach. Intell.* **34**(11), 2274–2282. https://doi.org/10.1109/TPAMI.2012.120 (2012).
38. Cho, K., Merriënboer, B.v., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H. & Bengio, Y. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In Alessandro Moschitti, Bo Pang, and Walter Daelemans, editors, *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1724–1734, Doha, Qatar, (2014). Association for Computational Linguistics. https://doi.org/10.3115/v1/D14-1179. https://aclanthology.org/D14-1179/.
39. Sundararajan, M., Taly, A. & Yan, Q. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ICML'17, p. 3319–3328. JMLR.org, (2017).
40. Walt, S.v., Schönberger, J.L., Nunez-Iglesias, J., Boulogne, F., Warner, J.D., Yager, N., Gouillart, E. & Yu, T. scikit-image: image processing in python. *PeerJ*, 2:e453 (2014) https://doi.org/10.7717/peerj.453
41. Chen, Z., Wang, Z., Yang, Y. & Gao, J. Resgraphnet: Graphsage with embedded residual module for prediction of global monthly mean temperature. *Artificial Intelligence in Geosciences*, **3** 148–156, (2022). https://doi.org/10.1016/j.aiig.2022.11.001. https://www.sciencedirect.com/science/article/pii/S2666544122000314.
42. Rozemberczki, B., Scherer, P., He, Y., Panagopoulos, G., Riedel, A., Astefanoaei, M., Kiss, O., Beres, F., Lopez, G., Collignon, N. & Sarkar, R. PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management*, 4564–4573 (2021).
43. Akiba, T., Sano, S., Yanase, T., Ohta, T & Koyama, M. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, KDD '19, 2623–2631, New York, NY, USA, (2019). Association for Computing Machinery. ISBN 9781450362016. https://doi.org/10.1145/3292500.3330701.

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

## Authors and Affiliations

**Tobias Milz[1] · Varvara Vetrova[2] · Marwan Katurji[1]**

✉ Tobias Milz
tobi.milz@canterbury.ac.nz

[1] School of Earth and Environment, University of Canterbury, Christchurch 8041, New Zealand

[2] School of Math and Statistics, University of Canterbury, Christchurch 8041, New Zealand